Managerial Flexibility and Project Valuation: Real Options

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Chapter Overview

This chapter considers situations where management has flexibility in the implementation of investment projects. Specifically, we consider cases where managers have flexibility regarding the timing of an investment's implementation as well as when it is shut down. In addition, we consider embedded real options that present themselves before an investment has been initiated. For example, an investment project can be designed to provide management with flexibility with respect to the products the firm produces or the inputs it uses. To value investments that benefit from managerial flexibility, we utilize (1) decision trees in combination with binomial lattices, (2) a real option pricing formula, and (3) simulation analysis.

1 Introduction

In the last half of the previous chapter, we evaluated an equity investment that could be characterized (and valued) as an option on a debt-financed investment. There, we learned that the option to default on the debt and walk away from the investment adds significant value to the equity investment. In this chapter, we extend our analysis to consider optionality that is *embedded* within the very nature of investment opportunities. These embedded options, which are generally referred to as *real options,* exist for any investment in which managers have the ability to exercise discretion. This discretion may involve timing issues such as determining when to initiate a new project or shut down an old one. In addition, managerial discretion can be exercised over how an existing investment is operated. For example, managers can respond to changing product demand by altering the mix of products and can respond to changing input prices by changing the mix of inputs they use. They can also speed up or slow down production of existing products, decide to extend existing product lines to incorporate new products, and implement a myriad of other choices that enable the firm to adapt to changing circumstances. Consequently, when we value real options, we are in a sense determining the value that an *active management team* can add when it responds in a dynamic way to a changing environment.

Consider, for example, an investment in a copper-mining operation. An important feature of copper ore is that it does not spoil or otherwise lose value when it is left in the ground. Consequently, if the price of copper drops below the cost of extracting, processing, and shipping the ore, management may choose to slow down production or temporarily shut down operations until market conditions improve. Similarly, if a real estate developer purchases land that is well suited for developing office property and the market for office space weakens due to a slowdown in the local economy, the developer may decide to delay the build-out of the property until the market begins to turn around or may consider other uses for the property. In both these examples, a key contributor to investment value is the embedded *flexibility* of the investment opportunities.

In the first half of this chapter, we will continue to illustrate concepts relating to real options within the context of oil field examples, which build on the ideas developed in the last chapter. For example, we will consider an oil field that provides its owner with the option to extract oil if oil prices are high and to keep the oil in the ground if oil prices are low. We will also continue to apply the binomial valuation model introduced in the last chapter.

In more realistic settings, the relevant options can be exercised at multiple dates, making their valuation quite complicated. In these cases, the valuation problem resembles the problem of valuing an American-style stock option, which is an option that can be exercised at any time prior to its expiration date. However, these real options are more difficult to value than stock options, because the underlying assets have uncertain payouts (analogous to dividends on a stock) and, in general, the exercise date is uncertain.

We demonstrate three approaches to solving the kinds of real option problems that are likely to arise in practice. The first is the binomial lattice, which is a multiperiod extension of the binomial option pricing model. We apply this approach to value an oil investment. The second approach applies an option pricing formula developed for use in valuing American call options that have infinite lives. We illustrate the use of this formula to value a real estate investment and a chemical plant. Finally, we use simulation to analyze operating options that arise when producers have the flexibility to choose between multiple modes of operation.

Because the analysis of real options is relatively new, we are aware of a number of different mistakes that are made in its application. To help our readers avoid these mistakes, we close our discussion of real options by identifying some common mistakes that analysts often make. Many of these errors arise out of the differences between options on financial assets (i.e., securities) and options on real investments.

2 Types of Real Options

Real options contribute to an investment's value whenever the following two conditions hold: (1) the environment is uncertain, and (2) managers can respond to changing circumstances by altering the way in which the investment is implemented and/or managed. It is useful to think about real options in terms of choices that are made *before* an investment has been launched and choices that are available to the managers who oversee the operations of an *ongoing business venture.* In either case, the availability of embedded optionality provides management with opportunities to exercise discretion that can have an important effect on the value of the project.

Real Options to Consider Before an Investment Launch

Before an investment is launched, the firm's focus is generally on timing issues and design-for-flexibility issues. The company grapples with questions such as the following:

Staged-investment options. In most cases, investments are made in stages. As we discussed, venture capital investments are generally made in stages. For example, a pharmaceutical start-up may invest in research in the first stage, which provides the firm with *an option* to develop and market a new drug if the research turns out favorably and if market conditions are promising. Another example is the acquisition of undeveloped land, which gives the owner *the option* to construct an office building, condos, or whatever is appropriate given future market conditions and zoning restrictions. Similarly, the acquisition of an undeveloped oil field gives the owner *the option* to develop the field and extract the oil. In each of these examples, we can think of the initial investment in the project as the *cost of a call option to invest in the next stage.*

Timing options. *Timing options* arise when it is possible to postpone the implementation date of an investment. The benefit of deferring an investment comes from the fact that the value of the investment is uncertain and some of this uncertainty will be resolved in the future.

Operating options. The fundamental problem here is how best to design or structure an investment so that it provides the firm with the operating flexibility necessary to respond to changes in the environment. The investment is more valuable if it is designed in a way that allows management to make choices that can capitalize on changing circumstances and the opportunities they bring.

Real Options to Consider After an Investment Launch

After an investment has been launched, the firm generally faces a different set of issues. In general, we refer to these as *operational issues,* and they too give rise to optionality that can enhance the value of the investment. Some examples where real options contribute to value include the following:

Growth options. This type of option includes the opportunity to expand both the scale and the scope of an investment. Expanding the *scale* of the investment refers to growing the specific project output through increased volume of production. Growing the *scope* of an investment includes such things as follow-on projects. These options are sometimes called *strategic options.* They are often used to justify investing in what appear to be negative-NPV projects if the projects open the door for a sequence of future investments.

Shutdown options. Almost all businesses have both good and bad times. Clearly, the business will be more valuable if management has the flexibility to shut it down during bad times and to operate it when the business is profitable.

Abandonment options. If a business is currently unprofitable and is expected to remain unprofitable in the future, it may make sense to abandon or sell it. The decision to abandon is influenced not only by the current profitability of the investment but also by the amount the firm could get for the business if it were to be sold.

Switching options—outputs. The ability to vary the output mix to reflect the relative value from alternatives can be a very valuable source of managerial flexibility.

Switching options—inputs. The ability to switch between two or more inputs provides managers with the opportunity to minimize input costs.

Even a cursory examination of the list of real options enumerated above suggests that most real investments contain some degree of optionality or discretion for the managers who design, implement, and operate the investment. The simple truth is that most investment projects are not the static opportunities that are frequently characterized in traditional discounted cash flow analysis; it is therefore critical that when the investments are valued, consideration be given to the inherent flexibilities that each investment opportunity offers.

3 Why Real Option Valuation Is Difficult

Real options, in general, are much more difficult to value than stock options and other options on financial claims. Perhaps the most obvious difference between real and financial options is that real option values are often determined by the values of other assets that are not actively traded. For example, in contrast to a stock—which might be traded every minute—factories, buildings, and other real assets are bought and sold very rarely. This fact makes it difficult to estimate how their returns are distributed and makes it impossible to hedge the risk associated with the real option by buying or selling the underlying asset.

A second difference between real and financial options is that the exercise choices are much more complicated with real options. With a stock option, one simply decides whether to buy the stock at some prespecified price. For real options, the decisions can be much more complicated. For example, a landowner must decide more than just whether to build on his land. He must decide what to build and how quickly to construct the building. In addition, he must consider the effect of changing construction costs (that is, an uncertain exercise price) and uncertainty about the value of the building. For example, the developer may want to design the building so that it can be more easily expanded to take advantage of the possibility of higher rents in the future.

4 Valuing Investments That Contain Embedded Real Options

This section opens our discussion of real option analysis with some simple examples that provide a framework for analyzing the importance of real options to project valuation. We then move progressively toward more-complex examples that better illustrate how these valuation problems are solved in practice. Where it is possible to do so, we utilize a valuation procedure; that is, we combine traded options into a tracking portfolio that mimics the payout of the real investment. In most cases, however, this is not possible and we will need to develop an approach to value the embedded options directly.

PRACTITIONER INSIGHT

Applications of Option Pricing Methods to Energy Investments—A Conversation with Vince Kaminski, Ph.D.*

We use option pricing methods extensively in the energy industry to value physical assets and long-term contracts. In general, any source of flexibility in quality, time of delivery, and so forth can be modeled as an option. Moreover, every source of rigidity in a contract contains embedded option components that can be modeled as interactive, multilayer options.

Traditional valuation methods for energy investments depend on estimates of future prices, and these forecasts are subject to the inherent limitations and biases of the individuals making the projections. Consequently, one advantage of the option valuation approach is that it relies on market prices that reflect the consensus across a wide range of traders. A second advantage of the option approach is that it relies on mathematical algorithms that were developed for use in valuing financial derivatives.

A very common example of the application of option valuation methods to physical assets is the valuation of a natural gas power plant. This investment can be viewed as a portfolio of short-term spread options on the difference between electricity prices and natural gas prices (adjusted for thermal efficiency, commonly referred to as "the heat rate" of the plant). However, there are complications that have to be incorporated into the modeling of such a plant. The plant operator faces physical constraints in the form of start-up costs, ramp-up and ramp-down costs, and added maintenance resulting from the wear and tear associated with switching the plant on and off in response to the demand for electricity. If these costs are ignored, then the estimated value of the plant (i.e., the spread options) will be biased upwards.

Yet another example of the application of option valuation methods in the energy industry involves the valuation of natural gas storage facilities. Depleted oil fields are often used to store natural gas produced during the summer months so that it can be sold during the winter months when the price of gas is higher. In this instance, the gas storage facility creates value as a calendar spread option. Salt domes can serve the same function. However, because the gas can be refilled and extracted from the salt dome much more quickly than from a depleted oil field, this type of storage facility provides the opportunity to respond much more frequently to short-term gas price spikes.[1] Consequently, this latter type of storage facility is often referred to as a "peaker storage facility."

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The Option to Invest: Staged Investments

In this section, we use an oil field investment example to illustrate the use of options for valuing a staged investment. In this particular case, the firm acquires an oil lease in the first stage that provides it with the opportunity to extract oil in later stages, depending on market conditions. To illustrate the value of this option to extract oil now or at a later

[1] Although these salt dome facilities can be abandoned salt mines, they often are not. After locating a salt deposit within a layer of relatively impervious rock, a hole is drilled into the formation, water is used to dissolve the salt, and the solution is pumped out to form the storage facility.

date, we consider the investment opportunity faced by Master Drilling Company, which is considering the purchase of an oil lease costing $450,000. The lease provides Master Drilling Company with the opportunity to develop oil reserves on a specified piece of property during the next year (later we relax this assumption and use a two-year lease). Because this particular lease is on property adjacent to a producing oil field, the company geologists are very confident about the quantity of oil that will be produced. Consequently, the primary concern that Master Drilling's management has about the venture relates to the price at which the oil will be sold.

Did you know?

Texas Hold 'Em Is a Game of Options

A great analogy can be drawn between the game of Texas Hold 'Em and real options. To see the connection, let's review how the game is played. Initially, players receive two down cards as their personal hand (hole cards), after which there is a round of betting. Each player must decide whether to match the highest bet, which buys her the option to continue to play or fold at the next round of betting. If a player decides not to bet, then she is choosing to abandon the hand. Next, three board cards are turned simultaneously (called the "flop") and another round of betting occurs, followed by two more board cards turned one at a time, with a round of betting after each card. Each round of betting provides the players with the opportunity to match or raise the highest bet or to fold. The winner is the player who forms the highest-ranking five-card hand using the hole cards plus community cards. Clearly, Texas Hold 'Em is a game of deciding when to acquire the option to stay in the game (betting) and when to exercise the abandonment option (folding).[2]

Valuing an Oil Lease

To demonstrate the value of the option to delay the decision to develop the property, we will initially assume that by purchasing the lease, the company is committed to start developing immediately and will extract the oil the following year. This requires an initial expenditure of $300,000 and an expenditure of $45 per barrel for each of the 100,000 barrels of oil extracted the following year. The following table summarizes the investment opportunity:

Today (Year 0)	**Year 1**
Master Drilling	Master Drilling
a. Purchases the lease for $450,000	**a.** Produces 100,000 barrels of oil at a cost of $45 per barrel
b. Spends $300,000 to develop the property in preparation to extract the oil in one year	**b.** Sells the oil at the prevailing market price

In addition, we make the following assumptions concerning the Master Drilling Company lease:

- The current forward price for the delivery of oil in one year is $50/barrel.
- An option to deliver a barrel of oil in one year with an exercise price of $45 per barrel can be bought or sold today for $8.50 per barrel.
- The risk-free rate of interest is 5%.

Traditional (static) DCF analysis of the lease To evaluate the decision to acquire the lease (assuming that Master Drilling begins development immediately and extracts and

[2] Oddly enough, this notion of options was immortalized in the chorus of the country hit "The Gambler" by Kenny Rogers when he sang,

> You got to know when to hold 'em, know when to fold 'em,
> Know when to walk away and know when to run.

sells the 100,000 barrels of oil in one year) using a traditional DCF analysis, we compute the NPV of the investment, as expressed in Equation 1 below, as follows:

$$NPV_{\text{Oil Lease}} = \frac{\left(\begin{matrix}\text{Expected Price}\\ \text{of Oil per Barrel}_{\text{Year 1}}\end{matrix} - \begin{matrix}\text{Extraction Cost}\\ \text{per Barrel of Oil}\end{matrix}\right) \times \begin{matrix}\text{Barrels of}\\ \text{Oil Produced}_{\text{Year 1}}\end{matrix}}{\left(1 + \begin{matrix}\text{Risk-Adjusted}\\ \text{Discount Rate}\end{matrix}\right)} - \left(\begin{matrix}\text{Cost of Acquiring}\\ \text{and Developing the Oil Lease}\end{matrix}\right) \quad (1)$$

Substituting for the values and estimates that we already know leaves the following:

$$NPV_{\text{Oil Lease}} = \frac{\left(\begin{matrix}\text{Expected Price of}\\ \text{Oil per Barrel}_{\text{Year 1}}\end{matrix} - \$45.00\right) \times 100{,}000 \text{ barrels}}{\left(1 + \begin{matrix}\text{Risk-Adjusted}\\ \text{Discount Rate}\end{matrix}\right)} - \$750{,}000$$

Thus, to solve for the project's NPV, we need to estimate an expected price for crude oil one year hence and also a risk-adjusted discount rate that is appropriate to the risks of developing the oil lease.

Certainty-equivalent analysis—using the forward price to value the lease The lease can also be valued with the certainty-equivalent approach using the observed forward price of oil. Specifically, for the price of oil we substitute the forward price, which is also the certainty-equivalent price. Because the resulting cash flow is now the certainly equivalent of the risky future cash flow, we can calculate its present value by discounting with the risk-free rate, as shown in Equation 2 below, i.e.,

$$NPV_{\text{Oil Lease}} = \frac{\left(\begin{matrix}\text{Forward Price}\\ \text{of Oil per Barrel}_{\text{Year 1}}\end{matrix} - \begin{matrix}\text{Extraction Cost}\\ \text{per Barrel of Oil}\end{matrix}\right) \times \begin{matrix}\text{Barrels of}\\ \text{Oil Produced}_{\text{Year 1}}\end{matrix}}{\left(1 + \begin{matrix}\text{Risk-Free}\\ \text{Rate}\end{matrix}\right)} - \left(\begin{matrix}\text{Cost of Acquiring}\\ \text{and Developing the Oil Lease}\end{matrix}\right) \quad (2)$$

Substituting for the forward price of oil in one year and the risk-free rate of interest, we calculate the NPV of developing the oil lease using Equation 2 as follows:

$$NPV_{\text{Oil Lease}} = \frac{(\$50.00 - 45.00) \times 100{,}000}{(1 + .05)} - \$750{,}000$$

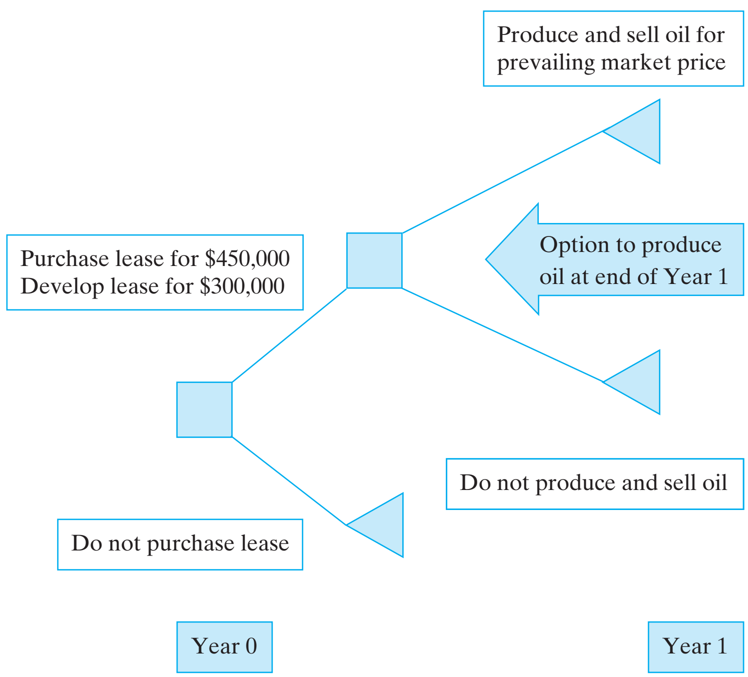
$$= \$476{,}191 - 750{,}000 = (\$273{,}810)$$

Clearly, under the assumption that Master Drilling must begin immediate development and commits to the production and sale of the oil reserves at the end of Year 1, the lease has a negative NPV! However, the typical lease *does not* commit the acquirer of the lease to produce and sell the oil reserves, but gives the buyer the right, not the obligation, to do so. In other words, the lease can be viewed as a call option on 100,000 barrels of crude oil that can be produced at a cost of $45 per barrel and sold one year hence.

Dynamic analysis—valuing the lease as an option to produce oil Up to now, we have assumed that when Master Drilling Company makes the investment to acquire the lease, it is obligated to develop and produce the reserves in the oil field. As the decision tree in Figure 1 illustrates, however, the firm has the option, but not the obligation, to produce the oil reserves and will extract the oil *only* if the revenue from selling the oil exceeds the \$45-per-barrel cost of extraction; otherwise, it will let the lease expire.

We assumed earlier that an oil option with a \$45-per-barrel exercise price currently has a price of \$8.50 per barrel. Hence, the lease should have the same value as 100,000 of these call options, or \$850,000. The relatively high price of the oil options in the derivatives market suggests that oil prices are relatively uncertain, and having the option to walk away from the lease without producing, or producing only when the price of oil exceeds extraction costs, is quite valuable.[3] This value exceeds the sum of the \$450,000 acquisition

Figure 1 **Dynamic Analysis—Oil Lease Investment As a Call Option**



[3] Using the binomial option pricing relation, we can evaluate just how volatile the future price of oil must be to warrant an option price of \$8.50 per barrel where the exercise price is \$45 per barrel. For example, assume that the price of oil in one year can be one of two prices: a high price of \$62.85 per barrel or a low price of \$37.15 per barrel. Using the forward price of \$50, we can calculate the implied risk-neutral probabilities of the two prices to both be 50% (i.e., \$50 = \$62.85 × Risk-Neutral Probability of the High Price + \$37.15 × Risk-Neutral Probability of the Low Price). The payoff to an option for one barrel of oil with an exercise price of \$45, then, is either \$62.85 − 45 = \$17.85 for the high price or \$0 if the low price occurs. We calculate the value of the option today as the risk-neutral probability-weighted average payoff to the option (i.e., its certainty-equivalent payoff), discounted at the risk-free rate of interest, i.e.,

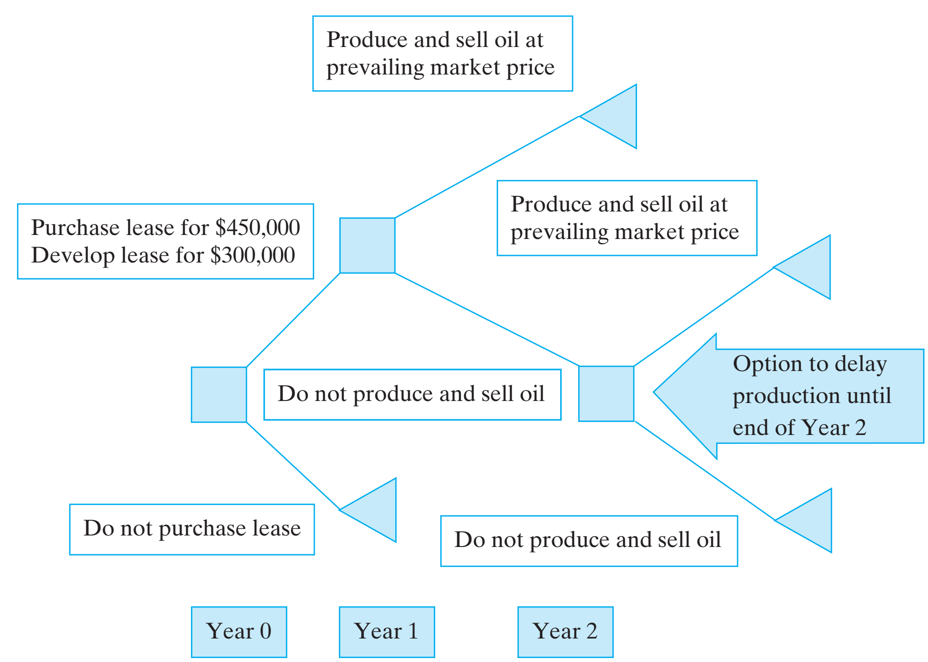
$$\begin{matrix}\text{Call Option} \\ \text{Value}_{\text{Today}}\end{matrix} = \frac{\$17.85 \times .50 + \$0.00 \times (1 - .50)}{(1 + .05)} = \$8.50$$

price of the lease plus the initial \$300,000 development costs, indicating that acquiring the lease and developing the oil field for possible production in one year (at a total cost of \$750,000) is a positive-NPV investment (i.e., NPV = \$850,000 − 750,000 = \$100,000).

Valuing the option to delay or wait In our preceding example, we analyzed the value of the lease as an option to extract oil either in the following year or not at all. However, like many other investments, oil leases also contain timing options that allow the owner to choose *when* to extract the oil. To illustrate the timing option, we continue with the Master Drilling Company example, but with a minor twist. We now assume that the lease allows Master Drilling two years to develop and produce the oil and that the company is now sitting at the end of the first year after having purchased the lease and spent the \$300,000 required to develop the property. In other words, the company is prepared to produce the oil at the end of Year 1 and must decide whether to do so now or wait a year until the end of Year 2. The revised investment opportunity is illustrated in Figure 2, where the company now has three decisions to make, as indicated by the decision nodes (squares) in the figure: (1) Should the company purchase the lease or not in Year 0? (2) If the answer to the first question is yes, then should the company produce and sell the oil reserves at the end of Year 1 or wait until the end of Year 2? (3) If the answer to the second question is to wait until Year 2 (i.e., do not produce in Year 1), then should the company produce and sell the oil at the end of Year 2 or abandon the lease?

Let's assume that the observed (spot) price of oil at the end of Year 1 is \$50/barrel, which exceeds the \$45 cost of extracting the oil. Consequently, the option to drill is *in the money.* However, Master Drilling's management now realizes that they have an additional option. They can extract the oil today or wait one more year and consider extracting the

Figure 2 **Oil Lease Investment with the Option to Delay**



oil then. If they do not extract the oil at the end of this second year, the lease will expire worthless. In addition to assuming that the price of oil is now \$50/barrel, we will assume that the one-year forward price (for Year 2) is also \$50 and that the value of an option to acquire oil one year hence (in Year 2), with a \$45 exercise price, still sells for \$8.50/barrel.

Because the price of oil at the end of Year 1 is \$50/barrel, immediately extracting the oil generates \$5 million in oil revenues. Subtracting the extraction costs of \$4.5 million = \$45/barrel × 100,000 barrels of oil generates a net profit of \$500,000. Note that this calculation does not consider the \$450,000 cost of acquiring the lease or the \$300,000 spent to develop the oil field and prepare it for production. These expenditures were made in Year 0 and as such represent sunk costs that do not influence the extraction decision. Consequently, the relevant comparison is between the \$500,000 cash flow from extracting the oil today and the discounted value of the expected cash flow from waiting and having the option to extract the oil at the end of Year 2.

Let's first ignore the option associated with the Year 2 extraction choice and consider the certainty-equivalent profits if the owner extracts the oil at the end of Year 2 of the original lease contract. Because the forward price is \$50/barrel for Year 2, the certainty-equivalent profit one year hence (at the end of Year 2) is the same as it was at the end of Year 1, or \$500,000, which has a present value (discounted at the risk-free rate) of \$476,191 at the end of Year 1. In this particular case, because the spot price for Year 1 and the forward price for Year 2 are equal (both are \$50 per barrel), the difference between the value of extracting now and extracting in the future arises solely because of the *time value of money*.[4]

Let's now consider how uncertainty and the flexibility to pass up the investment in the future affect the incentive to wait. As we have already shown, if an option to buy oil in the financial markets is valued at \$8.50 per barrel, and if we extract the oil in one year *only when oil prices exceed extraction costs,* the oil lease is worth \$850,000. This exceeds the \$500,000 value realized from extracting the oil immediately. Hence, the gains from waiting to develop the field only when oil prices are high more than offsets the effect of the time value of money, which makes it optimal to wait before investing.

Hedging the Price Risk of Delaying the Decision to Invest

You may find it counterintuitive to wait to invest when the environment is very uncertain. After all, if you delay the decision to drill, you face the risk of a declining oil price that can potentially make your lease expire worthless. Would it, instead, make sense to take a sure positive-NPV investment today rather than wait and hope for a higher-NPV investment in the future?

This intuition may have merit when the project's risks cannot be hedged. When no hedge is available, the investor's risk aversion may be an important factor in the decision problem.[5] However, if the Master Drilling Company can sell call options on oil with

[4] It should also be noted that the slope of the forward curve also provides an incentive to either produce now or wait. If forward prices in the future are much higher than the current spot price, this offsets the effect of the time value of money and increases the incentive to delay production. For example, if the forward price of oil was \$55/barrel rather than \$50, it would clearly pay to wait.

[5] In theory, risk aversion should not enter the decision of a publicly traded firm. However, in reality, risk-averse managers may not want to delay positive-NPV investments that may need to be abandoned in the future. Also, Accounting issues may influence the decision to delay an investment.

Table 1 Hedging the Option to Delay with Call Options

	Price of Oil	
	$45.00 or below	**$ 60.00**
Project Earnings per Barrel (= Price/Barrel − Extraction Costs)	$ 0.00	$ 15.00
Minus Call Payoff per Barrel (= Max[Price/Barrel − $45, 0])	$ 0.00	$(15.00)
Call Price per Barrel	$ 8.50	$ 8.50
Total	$ 8.50	$ 8.50

a strike price of $45 per barrel, it can hedge these risks and lock in the gains associated with waiting to invest. Table 1 describes the payoffs per barrel of oil associated with maintaining the option to extract the oil in the following year and hedging the uncertainty by selling call options on 100,000 barrels of oil with an exercise price of $45 per barrel at the current market price of $8.50 per barrel. As the numbers in the table indicate, the waiting-and-hedging strategy locks in a gain of $8.50 per barrel, which exceeds the gain from extracting the oil immediately.

If the price of oil drops below $45 per barrel, then both the project and the call options that the firm sold are worthless. Similarly, if the price of oil were to rise to $60 per barrel, then the project payoff is $60 per barrel − $45 per barrel = $15 per barrel, which again is exactly equal to the payout Master Drilling must pay to the holders of the call options it sold. In both cases, the payoff from the hedged position is simply the $8.50-per-barrel proceeds from the sale of the options.

More Complicated Options and the Incentive to Wait

The preceding example indicates that price uncertainty provides an incentive for firms to delay their investments. This example is easy to evaluate because the risks associated with oil price uncertainty can be hedged, providing the firm with the opportunity to lock in the gains associated with waiting. However, the option to delay should be viewed broadly, and one should not conclude that realizing the value to delay requires that the risks be hedged. The general principle, that the value of the option to delay comes from the opportunity to learn more about the investment's prospects, can be applied to almost any investment. By learning more, the firm may decide to configure the investment in a different way or to simply expand or contract the scale of the investment in light of what it learns during the delay.

One straightforward example arises in the context of undeveloped land, which can be viewed as an option to acquire a building at the cost of construction. When there is only one type of building that can be built, the option is a simple call option, just like the oil lease. However, the owner of the land may have the opportunity to build a number of different types of structures on the land, such as condos, a hotel, or an office building. When this is the case, the investor's options are somewhat more complicated, making them more difficult to value and hedge. However, the intuition is still the same: Greater uncertainty makes the options more valuable and increases the incentives to wait to invest. In general, *when the options provide more flexibility, they are more valuable and the benefits from waiting are greater.* Intuitively, when you have more choices, it pays to wait to see how uncertainty unfolds to make sure that you are making the best choice.

Do Firms Delay Optimally?

For a variety of reasons, managers may not be as willing to delay the initiation of positive-NPV investments as the real options model suggests they should be. Our introductory finance classes may be partially to blame, because they tend to place too much emphasis on the static NPV rule, which suggests that firms should undertake all projects with positive NPVs. However, these same textbooks also say that positive-NPV investments may be passed up when there exist other, *mutually exclusive* investments with even higher NPVs.

In essence, whenever an investor has discretion as to the timing of an investment, this creates mutually exclusive alternatives. For example, Project 1 is taking the project immediately, and Project 2 is waiting and taking the same project at a future date. It may be the case that Project 1 has a positive NPV but that Project 2, which requires waiting (and of course cannot be implemented if Project 1 has already been implemented), may have a higher NPV and would therefore be preferred.

We should also stress that this tendency to initiate positive-NPV investments too early relative to the specification of the real options model may not simply be a mistake. Managers may be reluctant to wait on a positive-NPV investment for a number of good reasons:

1. *In reality, we cannot perfectly hedge the real option by selling financial options.* As a result, waiting is risky.
2. *The investor may be credit constrained or may face very high borrowing costs.* Credit constraints can have the effect of speeding up some investments and slowing down others. For example, an investor may extract oil more quickly than he otherwise would if doing so provides capital that can be invested in positive-NPV investments that could not be funded otherwise. However, if speeding up extraction requires new capital, then having limited access to financial markets to raise the capital can delay the investment.
3. *Initiating the project may generate information that would be useful to the firm's managers in evaluating future investments.* For example, the quantity of oil that is extracted may provide information about the potential amount of oil in similar regions. In this instance, the information gleaned from early exercise provides the firm with valuable information about future prospects. In essence, exercising one option may increase the value of a firm's other options.
4. *It may be important to signal to outside investors the fact that the firm does indeed have a positive-NPV investment.* If outside investors are not as well informed as the firm's management, it may be important that the firm periodically provide signals related to its ability to generate positive-NPV projects. For example, to show investors that they have the best geologists, an oil firm may need to develop some of its oil fields. Once again, early exercise may provide extra value and benefits to the firm and its future projects. If, for example, the firm is about to enter the capital markets to raise additional financing, revealing information about the prospects of new investments could lead to a higher stock price and a lower cost of capital. These benefits may overshadow the lost value from exercising early.
5. *Managers may undertake early initiation for personal reasons.* The compensation practices of firms that reward current-period performance can provide very strong incentives for managers to exercise their investment options too early. These practices tend to reward executives for immediate performance and penalize long-term investments that have too detrimental an impact on current-period performance.

Option to Abandon

Financial analysts have long recognized the value of the option to abandon an investment that is not performing as planned. For example, if you own a widget factory that is generating very low cash flows, you may be better off closing the factory and selling the property to a real estate developer who wants to convert the space for loft apartments. Moreover, if the cash flows are negative, you might want to abandon the factory even if the property has no alternative use.

The abandonment option has an important timing component that is very much in the same spirit as the "option to defer or delay" that we considered earlier. Just as option theory implies that there is an incentive to defer a positive-NPV investment, there may also be an incentive to continue operating a money-losing investment. Intuitively, the combination of uncertainty and flexibility should make decision makers more cautious about making irreversible decisions to either initiate a project or abandon a project. The key assumption that drives this conclusion is that the decision to permanently shut down or abandon an investment is irreversible.

Stripper Well Example

To illustrate the shutdown or abandonment option, we analyze the decision to shut down a **stripper well,** which is an oil well that produces relatively low volumes of oil each year and frequently is just barely profitable. Prior to 1998, there were a number of stripper wells in Texas that had annual production volumes of 1,000 barrels of oil or fewer. Because these wells produce very little, their operating costs per barrel of oil are relatively high, which means that although they can be quite profitable when oil prices are high, many operate at a substantial loss when oil prices are low.

At the end of 1998, oil prices fell to close to \$10 per barrel, making most of the Texas stripper wells unprofitable. Because of their operating losses, a number of the wells were closed in 1999, without the possibility of being reopened.[6] Oil prices subsequently increased substantially, making the stripper wells that had not been shut down quite profitable. With the benefit of hindsight, one can argue that the decision to shut down the wells at the end of 1998 was a mistake. What we need to consider here, however, is whether shutting down the wells was a wise, or at least a defensible, decision in 1999, given the information that was available at that time.

To illustrate how we might use real option analysis to evaluate the option to abandon these wells, we will consider a regional oil producer with 100 stripper wells that can produce 100,000 barrels both this year and next year before depleting their reserves. The cost of operating the wells and transporting the oil is \$14 per barrel, and the wells can be shut down either now or at the beginning of next year. The current price of oil is \$10 per barrel, and the forward price for delivery in one year is also \$10 per barrel. However, because oil prices are extremely volatile, an option to buy oil in one year for an exercise price of \$14 a barrel sells for \$4.50 today.

[6] The problem encountered in shutting down production is that the formation may collapse or otherwise stop the flow of oil. This, in turn, means that for practical purposes, the decision to shut down production is tantamount to abandoning the well.

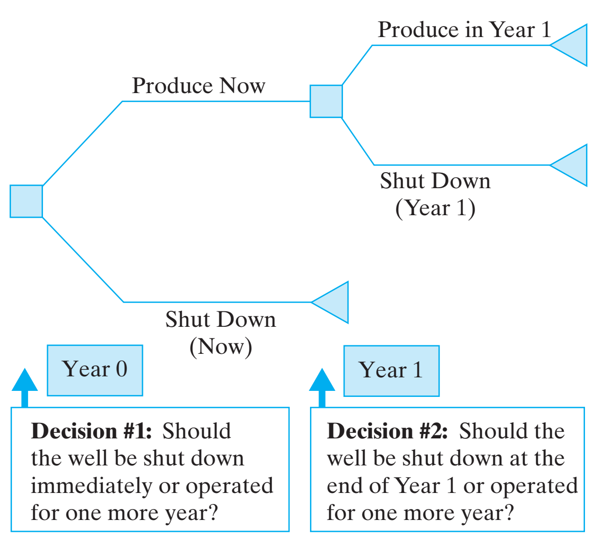
Given the current oil price and the forward price in one year, we can evaluate the current value of the 100 stripper wells, assuming that the oil producer will operate the wells this year and next year. We again assume a risk-free rate of 5%:

$$\begin{array}{c}\text{Present Value of}\\ \text{Stripper Well}\\ \text{Cash Flows}\end{array} = \underbrace{100{,}000\,\text{bbls} \times (\$10/\text{bbl} - 14/\text{bbl})}_{\substack{\text{Present Value of Operating}\\ \text{the Wells in Year 0}}} + \underbrace{\frac{100{,}000\,\text{bbls} \times (\$10/\text{bbl} - 14/\text{bbl})}{(1 + .05)}}_{\substack{\text{Present Value of Operating}\\ \text{the Wells in Year 1}}}$$

This analysis indicates that if the *only* alternative is to operate the well for the next two years, it is best to shut the wells down, because the investment has a large negative NPV. However, if the owner of the wells has the *option* to shut down the wells next year *if* oil prices continue to be low (illustrated in Figure 3), it may in fact make sense to continue operating the wells this year at a loss of ($400,000), because shutting them down now eliminates the option to produce next year. If the loss from *producing now* is less than the *value of the option to produce next year,* it makes sense to continue to operate the wells.

In this particular example, we see that an option to buy oil at $14 per barrel generates the same cash flow as the stripper wells in one year. If oil prices recover and rise above $14, the wells will produce per-barrel profits equal to the oil price minus the $14-per-barrel extraction costs. This net cash flow is exactly the same as the payoff on the option contract in this situation. If, however, oil prices stay below $14 per barrel, the options will expire worthless and the oil wells will be shut down and will also generate no revenue. Hence, the *real* option to extract oil in the next year has the same value as the financial option to purchase oil for $14 per barrel, which we earlier assumed could be bought and sold for $4.50 per barrel.

Figure 3 **Decision Tree for the Stripper Well Abandonment Problem**



In a sense, by continuing to produce this year, the investor acquires an option to produce next year. When prices are very volatile, having the right, but not the obligation, to produce in the following year can be very valuable, and this option may exceed the loss incurred by operating the wells in Year 1. So, what is the value of continuing to operate the stripper wells? The total value equals the sum of the profits from extracting the oil this year, which is a negative number, plus the value of the option to extract next year. Our analysis indicates that the stripper wells should not be shut down, because the loss suffered from operating the wells in the current year is more than offset by the value of the option to extract 100,000 barrels next year. That is:

$$\begin{array}{c}\text{Value of the}\\ \text{Stripper}\\ \text{Well}\end{array} = \left(\begin{array}{c}\text{Value}\\ \text{of Operating}\\ \text{in Year 0}\end{array}\right) + \left(\begin{array}{c}\text{Value of the}\\ \text{Option to Operate}\\ \text{in Year 1}\end{array}\right)$$

$$= 100{,}000 \text{ barrels}(\$10/\text{barrel} - \$14/\text{barrel}) + \$4.50 \times 100{,}000 \text{ barrels}$$

$$= (\$400{,}000) + \$450{,}000 = \$50{,}000$$

Hence, there is positive value from operating the stripper wells this year that is derived from the fact that by operating the wells, the firm obtains the option to operate them next year. Consequently, it is best not to shut down the wells.

Note that in the preceding example, we assumed that there was no cost of shutting down the stripper wells immediately, beyond the loss of value derived from the option to produce in the second year. In many types of investments, however, there *are* costs incurred to shut down or decommission an investment, and these costs are sometimes so large that they dominate the decision. For example, refineries and chemical plants can face large cleanup costs when they are shut down. These cleanup costs may be so massive that the opportunity to delay incurring them may be a primary reason for keeping an old plant operating.

5 Analyzing Real Options As American-Style Options

The previous discussion presented very simple examples that included, at most, two possible drilling or abandonment dates. Our objective in these examples was to develop a framework that would help build intuition for how analysts and firms solve these problems in practice. Unfortunately, there are no simple formulas (nor even any complicated ones) that we can use for realistic valuation problems involving real options. In this section, to provide more insight about how these problems are solved in practice, we introduce a bit more complexity by means of another oil example, this one being more realistic than the examples discussed previously.

Evaluating National Petroleum's Option to Drill

National Petroleum, a small exploration and production (E&P) company, has a lease that provides the company with an option to drill on the property within a period of three years. If National drills and finds oil, it has the option to extract oil on the property until the reservoir is depleted, which is estimated to be 10 years after drilling is initiated.

Figure 4 **Production Volume for the National Petroleum Lease**

Year 0	0 barrels
Year 1	400,000 barrels
Year 2	300,000 barrels
Year 3	200,000 barrels
Years 4–10	100,000 barrels

Specifically, the property is expected to produce oil at the rate described in Figure 4, where Year 0 is the drilling date. If, however, the company chooses not to drill within the allotted three-year period, the lease and the option to drill will expire.

Hedging Oil Price Risk

National Petroleum estimates the cost of drilling to be $38 million and the extraction and delivery costs to be $28 per barrel. Once again, our problem is to place a value on the lease. To establish a base value for the property, we assume that drilling begins immediately and that all of the production is sold forward at the forward prices found in Table 2. This strategy effectively hedges the price risk of the project cash flows and

Table 2 **Calculation of Project Revenues Using Forward Prices for Oil**

Year	Forward Price	Volume (barrels)	Revenue	Extraction Cost	Net Cash Flow
1	$59.00	400,000	$23,600,000	$(11,200,000)	$12,400,000
2	$60.00	300,000	18,000,000	(8,400,000)	9,600,000
3	$61.00	200,000	12,200,000	(5,600,000)	6,600,000
4	$62.00	100,000	6,200,000	(2,800,000)	3,400,000
5	$62.00	100,000	6,200,000	(2,800,000)	3,400,000
6	$63.00	100,000	6,300,000	(2,800,000)	3,500,000
7	$63.00	100,000	6,300,000	(2,800,000)	3,500,000
8	$63.00	100,000	6,300,000	(2,800,000)	3,500,000
9	$63.00	100,000	6,300,000	(2,800,000)	3,500,000
10	$63.00	100,000	6,300,000	(2,800,000)	3,500,000
PV (6%) =	$42,034,394				
Drilling cost =	(38,000,000)				
NPV =	$ 4,034,394				

locks in the profits based on these forward prices.[7] Because the resulting profits are now hedged (i.e., risk free), we can value the project by discounting the future cash flows using the risk-free rate of 6%.

Table 2 presents the hypothetical forward prices for the next 10 years, in addition to volumes, revenues, extraction costs, and the resulting free cash flows for the investment. We have evaluated the NPV of drilling today using the forward price curve to hedge oil price risk and found the NPV to be $4,034,394. However, National does not have to drill immediately, according to the terms of the lease. In fact, the forward price curve may be such that drilling at the end of Year 1 and producing in Years 2 through 11 might be even more valuable. Similarly, National could initiate drilling at the end of Year 2 or even Year 3. If we know the forward prices for oil in Years 11 through 13, we can use the same hedging strategy to value the opportunity to drill in each of these three years and then select the best time to drill. If the forward price curve remains flat at $63 a barrel in Years 11 through 13, then the decision today to commence drilling in each of the next three years produces the following NPVs:

Commence Drilling in Year	NPV (Present Values as of Year 0)
0	$4,034,394
1	$4,642,831
2	$5,024,001
3	**$5,197,453**

Clearly, given this forward price curve, National should *not* drill immediately, because the firm can lock in the highest value by committing to drill in three years and hedging the oil price risk. Note, however, that the above analysis *presumes* that National commits *today* to a drilling program that does not begin until the end of Year 3. In reality, National has the option to drill at the end of Year 2 if it chooses not to drill in Year 1, so we should view the NPV calculated above as a very conservative estimate of the value of the oil field. If we consider the fact that National has the option to defer the decision to initiate the drilling operation for as long as three years, the lease becomes even more valuable.

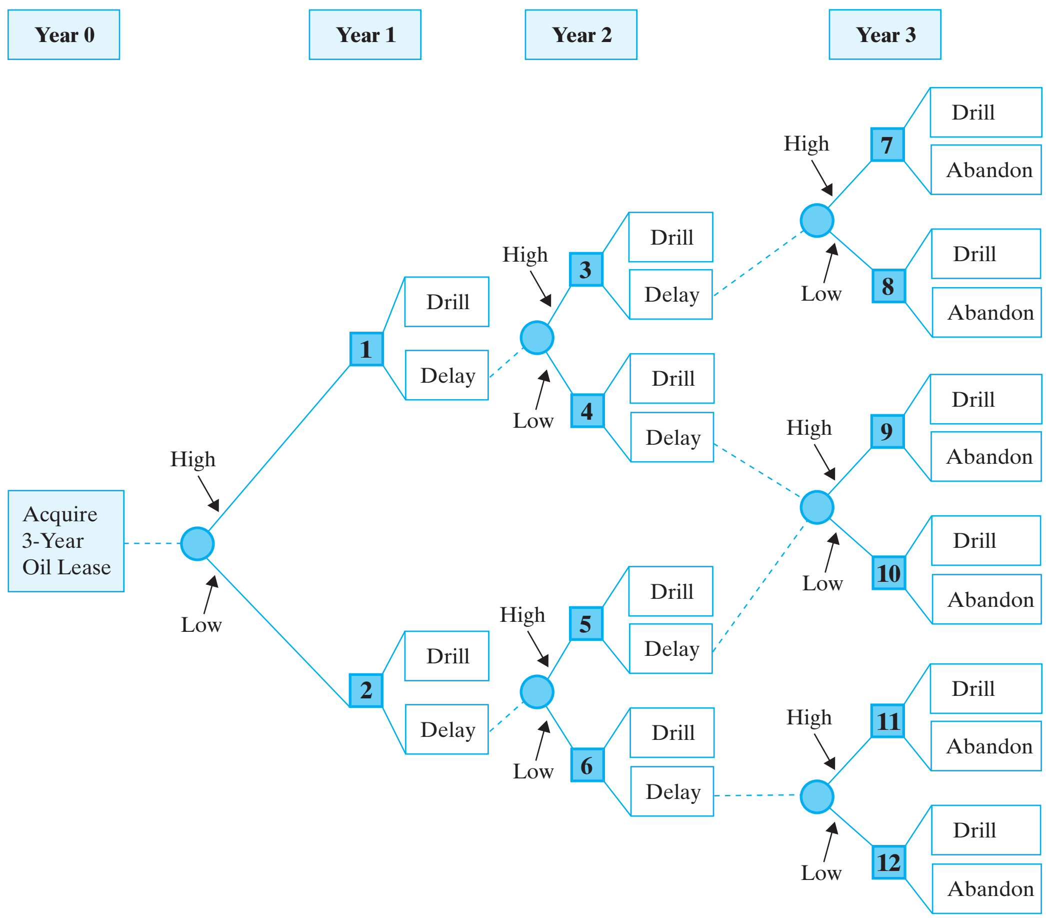
Considering the Option to Delay

Figure 5 uses a decision tree to illustrate the options available to National during the three-year period of the oil lease. This decision tree has as its basis the binomial model that specifies oil prices along with an illustration of the decisions that can be made. Recall that the circles indicate a random event (in this case, whether the price of oil will be the high price or the low price) and the squares indicate that a decision must be made. For example, assume that at the end of Year 1 the price of oil is equal to the high price.[8] National's management now faces decision node #1 and will have to decide whether it is best to drill now or delay until Year 2. If the decision is made to drill (i.e., exercise the option to drill), then that is the end of that branch of the

[7] Again, recall that this assumes that we know the quantity that will be produced. This is an important caveat, because profits depend upon both oil prices *and* the volume of oil produced. Thus, our ability to hedge the risk associated with our profits depends on both price and production volume.

[8] To simplify our exposition, we assume that decisions are made annually and that the time step used to define the binomial lattice is also one year.

Figure 5 **Decision Tree for the Option to Delay**



decision tree. However, if the decision is made to delay until Year 2, then we follow the dashed line to the appropriate probability node for Year 2, where we learn whether the price of oil will be high or low, leading to decision nodes #3 and #4, respectively. Note that if National has not exercised its option to develop the property by the end of the three-year lease term, it effectively abandons the investment opportunity.

To this point, we have not described how the decision should be made with regard to drilling now versus delaying the decision to drill. Incorporating the option to defer drilling for up to three years into our analysis requires that we evaluate the investment opportunity as follows: To determine whether the option to drill should be exercised, we compare the value of the oil field today, minus drilling and development costs, with the value of the option to drill and develop the well in the future. When the value of the option to drill in the future exceeds the value of drilling immediately, it is better to wait, and vice versa.

In this particular case, National Petroleum has the option to drill at any time up to the expiration of the three-year lease, which provides the firm with what is analogous to

Table 3 Comparing Real Options with American Stock Options

Inputs for Valuing American Stock Options	Oil and Gas Field Counterparts
1. Stock return volatilities	*Volatility* of the value of the producing field
2. Value of underlying stock	The present value of the cash flows from the producing wells
3. Dividend yield	The *convenience yield* is the benefit associated with the physical ownership of the commodity.*
4. Expiration date	The *expiration date* on the lease
5. Exercise price	The *cost of drilling*

*When the convenience yield is high, the benefit from having the commodity today is high, which means that the price today is high relative to the expected price in the future. The convenience yield is important because it determines the expected rate at which the commodity will appreciate and is related to the slope of the forward curve minus the risk-free rate.

an American call option on a stock. That is, National Petroleum has what is equivalent to an option to acquire a producing oil well at an exercise price that equals the cost of drilling. To value the lease, we would like to find the market price of a traded option that exactly matches the option that we are trying to value. However, because it is unlikely that there exists a publicly traded option on a producing oil well that exactly matches the option associated with this particular oil field, we will need a model to value the option. Such a model requires estimates of several key inputs, which are analogous to the inputs needed to value an American stock option on a share of stock. Table 3 summarizes the analogy between the American call option problem and the option to extract oil problem.

Valuing the option to drill is substantially more difficult than solving a typical American stock option valuation problem, though. This is true primarily because the input that determines the expected change in the value of the developed well—the *convenience yield* (the counterpart to the dividend yield)—changes from period to period as the prices in the forward curve change. In addition, the *volatility* of the returns to investing in the producing well is likely to change over time and is difficult to estimate.

Valuing American Options Using a Binomial Lattice

In practice, analysts frequently use a multiperiod version of the binomial option pricing model to evaluate American-style real options. To show how this works, we will apply such a model to the valuation of the drilling investment opportunity faced by the National Petroleum Company. Because National has the option to initiate drilling at any time until the end of a three-year period, the problem National faces is to value a three-year American call option.

Earlier, we estimated the value of the drilling opportunity to be $42,034,394 if drilling begins immediately. However, this value does not reflect the added value that might be captured if the firm exercises its option to delay drilling until a later period, when oil prices might be more favorable. To value this option, we start by constructing the multiperiod binomial lattice found in Figure 6. This lattice, which is based on the oil prices described in the decision tree in Figure 5, describes the possible values that the drilling opportunity might have in each of the next three years (the life of the American call option we wish to value).

We can think of the values at each node of the binomial lattice as the discounted values of future project cash flows based on the forward curve of oil prices as they evolve through time. The actual future values of the developed oil field that are specified in the binomial lattice can be somewhat arbitrary and may depend on the judgment of the decision maker. (We address this issue in more detail in the Appendix.) But first, let's walk through the valuation process *assuming* that we know the value of developing the oil field at each node of the binomial lattice.

Four values are listed at each node of the binomial lattice in Figure 6. We will refer to them as valuations #1 through #4, and they represent (from top to bottom):

> *Valuation* #1: *The value of the developed field*—the estimated value of initiating the development of the undeveloped oil field on this date, which is equal to the DCF valuation based on the forward price curve for this period and the assumption that drilling begins in this year and production follows in the subsequent 10-year period.
> *Valuation* #2: *The NPV of waiting to drill*—the expected value of the undeveloped field if the option to drill is not exercised this period.[9]
> *Valuation* #3: *The NPV of drilling now*—that is, the value of the oil field if developed this period (valuation #1 above) less the $38 million cost of drilling and developing the field.
> *Valuation* #4: *The value of the undeveloped oil field*—This valuation is analogous to that of an American call option, which is equal to the maximum of either the NPV of waiting until the next period to drill (valuation #2 above) or the NPV of drilling in the current period (valuation #3 above).

Solving for the value of the undeveloped oil field To value the option associated with owning the undeveloped field, we must work backward in the binomial lattice, beginning in Year 3 (the year in which the option expires). This procedure is commonly referred to as *rolling back the branches of the decision tree.* The analysis for Year 3 proceeds as follows: The value of the option to drill in Year 3 is the maximum of the NPV of drilling in Year 3 (i.e., the value of the developed oil field minus the $38 million development cost) or zero. For example, if the value of the developed field is $63.98 million in Year 3 (the highest-value node), then the value of the undeveloped oil field equals

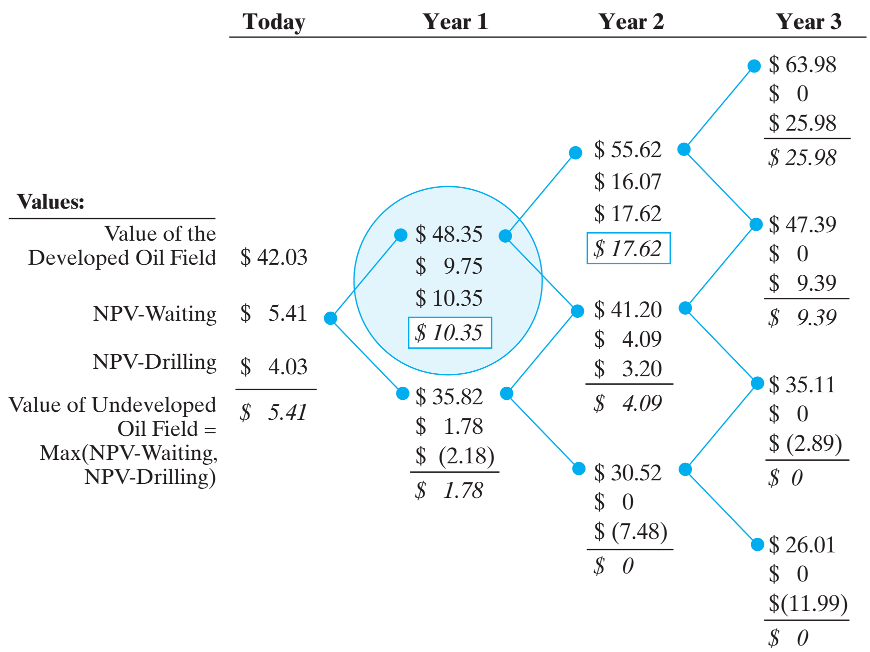
$$\text{Max}(\$63.98 \text{ million} - \$38 \text{ million}, 0) = \$25.98 \text{ million}$$

In this particular node, it is optimal to develop the field; however, this is not always the case. For example, if the value of the developed oil field in Year 3 were equal to the lowest node in the lattice, which is $26.01 million, the value of the undeveloped field is zero, because the NPV of drilling at this node is negative (i.e., $26.01 million − $38 million = −$11.99 million).

After evaluating all the Year 3 nodes of the lattice, we now consider the nodes for the end of Year 2. In the highest-value node for Year 2, the value of the developed oil

[9] For example, the NPV of waiting until the end of Year 1 is equal to the present value of the undeveloped oil field at the end of Year 1. That is, it is the present value of the probability-weighted values of the undeveloped oil field at the end of Year 1, which are equal to $10.35 million or $1.7822 million.

Figure 6 **Binomial Tree Describing the Possible Values of the Drilling Opportunity (in millions)**



Definitions

(a) *Value of the developed field* = present value of future cash flows from the oil field spanning a period of 10 years from the date of the valuation—for example, in the circled node at the end of Year 1, this value is equal to $48.35 million = the present value of the oil field if developed at the end of Year 1 and if evaluated using the forward price curve for oil produced in Years 2 through 11. Technically, this is the way to think about the value of the oil field if drilling and development commence during Year 1. However, the forward curve that will exist at the end of Year 1, where the high-price state occurs, is not observable today when we are performing our analysis. Consequently, we model the price distribution for the value of the oil field using the binomial process described in the Appendix to this chapter.

field is equal to $55.62 million. The NPV of waiting is equal to the present value of the certainty-equivalent payoff[10] from the option to begin drilling in Year 3, i.e.:

$$\text{NPV of Waiting to Drill} = e^{-.06}[\$25.98 \text{ million} \times .4626 + \$9.39 \text{ million} \times (1 - .4626)]$$
$$= \$16.07 \text{ million}$$

The NPV of drilling at the end of Year 2 is $17.62 million, calculated as the value of the developed property minus the $38 million drilling cost. Because the NPV of drilling

[10] The certainty-equivalent payoff from the option is equal to the weighted-average payoff where the weights are the risk-neutral probabilities. In the binomial lattice found in Figure 6, the risk-neutral probabilities of up and down prices are equal to .4626 and (1 − .4626), respectively.

(b) *Risk-neutral probabilities.* We have calculated the risk-neutral probability using the forward price of oil. This method can be translated into the following calculation to get the risk-neutral probability of the high-price state (P) using the following relationship:

$$P = \frac{e^{(r-\mathrm{d})} - e^{(r-\mathrm{d})-\mathrm{s}}}{e^{(r-\mathrm{d})+\mathrm{s}} - e^{(r-\mathrm{d})-\mathrm{s}}}$$

where r is the risk-free rate of interest, δ is the convenience yield, and σ is the volatility of the returns to the oil field investment. As discussed in the Appendix, we assume that $r = 6\%$, $\delta = 7\%$, and $\sigma = 15\%$. These values imply that $P = 0.4626$.

(c) *NPV of waiting* = present value of exercising the option to drill at a later date—for the circled node at the end of Year 1, this is equal to \$9.75 million = the present value of the risk-neutral probability-weighted expected value of the call option in Year 2, i.e.,

$$\begin{array}{c}\text{NPV of Waiting}\\ \text{to Drill}\end{array} = e^{-.06}[\$17.62 \text{ million} \times .4626 + \$4.09 \text{ million} \times (1 - .4626)]$$
$$= \$9.75 \text{ million}$$

Note that the payoff to the option to wait to drill (evaluated at circled node at the end of Year 1) is based on the maximum of the NPV of waiting until Year 2 to decide whether to drill or not for each potential node [i.e., for the high-price node, the payoff to waiting until Year 2 is max(\$16.71 million, \$17.62 million)]. For the low-price node, the payout to waiting is equal to max(\$4.09 million, \$3.2 million). The risk-neutral probability of the high-value state was defined in (b) above.

(d) *NPV of drilling now* = value of developed oil field cash flows if drilling takes place in the current period [item (a) above] minus the cost of drilling = \$10.35 million for the circled node in Year 1—calculated as follows: NPV of drilling now (circled node) = value of the developed field (circled node) — cost of developing the field, i.e.,

$$\$10.35 \text{ million} = \$48.35 \text{ million} - \$38 \text{ million}$$

(e) *Value of the undeveloped oil field* = the maximum of the NPV associated with waiting to exercise the option to drill [(c) above] and the NPV of exercising the option in the current period [(d) above]. This is analogous to the value of an American call option on the value of developing the oil field—\$10.35 million = maximum of the NPV of waiting and the NPV of drilling now:

$$\begin{array}{c}\text{Value of the}\\ \text{Undeveloped}\\ \text{Oil Field}\end{array} = \text{Max}(\{e^{-.06}[\$17.62 \text{ million} \times .4626 + \$4.09 \text{ million} \times (1 - .4626)]\}, \{\$48.35 \text{ million} - \$38 \text{ million}\})$$
$$= \text{Max}(\$9.75 \text{ million}, \$10.35 \text{ million}) = \$10.35 \text{ million}$$

exceeds the NPV of waiting to drill, the optimal decision is to drill if this node in the lattice occurs. Hence, the value of the undeveloped oil field is defined as follows:

$$\begin{array}{c}\text{Value of the}\\ \text{Undeveloped}\\ \text{Oil Field}\end{array} = \text{Max(NPV of Exercising Now, NPV of Waiting to Exercise)}$$

where "exercising" refers to the initiation of drilling and production of the field. Substituting for the appropriate values needed to evaluate the option in the highest-value node for Year 2,

$$\begin{array}{c}\text{Value of the}\\ \text{Undeveloped}\\ \text{Oil Field}\end{array} = \text{Max}\{\$55.62 \text{ million} - \$38 \text{ million}, e^{-.06}[\$25.98 \text{ million} \times .4626 + \$9.39 \text{ million} \times (1 - .4626)]$$
$$= \text{Max}(\$17.62 \text{ million}, \$16.07 \text{ million}) = \$17.62 \text{ million}$$

Note that when we evaluate the option to wait at the end of Year 1, the value that we use for Year 2 to calculate the certainty-equivalent value is the maximum of the NPV of the field if it is developed in Year 2 (i.e., $17.62 million in the example calculation above) and the value of waiting until Year 3. The latter is the present value of the expected NPV of drilling or waiting for each of the two value nodes connected to the top node in Year 2. That is, for the high-value node in Year 3, we have already determined that the maximum of the NPVs of drilling and waiting (abandoning the opportunity because the lease expires) is to drill, for an NPV of $25.98 million. Similarly, if the next-lower node (the low-value state compared to the high-value node in Year 2 that we are analyzing) occurs, then the maximum NPV for the investment entails drilling, and this opportunity has an NPV of $9.39 million.

We now have all the information we need to determine the optimal exercise strategy for developing the three-year lease of the oil field. Specifically, reviewing the completed binomial lattice found in Figure 6, we see that exercising the option immediately in Year 0 has a positive NPV of $4.03 million = $42.03 million – $38 million. The value of waiting to implement the investment at a later date is $5.41 million. Therefore, National should not begin drilling immediately but should wait until the end of Year 1 and reevaluate. If at the end of Year 1 the undeveloped oil field is worth $48.35 million, it will be better to exercise the option to drill and develop the property. Put another way, the value of waiting until the end of Year 2 is worth only $9.75 million, while the NPV of developing the oil and gas reserves at the end of Year 1 is $10.35 million. If, however, at the end of Year 1 the value of the undeveloped oil field is equal to $35.82 million (the lower node in Year 1), then the field should not be developed. This is because the NPV of waiting is worth $1.78 million, while the NPV realized by implementing the development process at the end of Year 1 is negative (i.e., −$2.18 million).

Constructing the binomial lattice for the developed oil field In Figure 6 in the preceding example, we provided the valuations of the oil field investment in each node at the end of Years 1, 2, and 3. In practice, the analyst would not be given these values but would have to estimate them. Because this estimation process, which is referred to as "calibrating the model," is quite complex, we will briefly summarize how it is done in practice. (Calibration is discussed in more detail in the Appendix at the end of this chapter.) We will not delve into the specifics of the calibration process here, for that is where much of the "heavy lifting" occurs in modeling commodity prices, and it can be quite technical.

The first step in the estimation process is to specify a binomial tree that illustrates the distribution of future oil prices. Because it takes 10 years to produce the oil field's reserves and production may not commence for three years, we need to specify the distribution of crude oil prices for the next 13 years. The price distribution should be chosen so that the expected future prices at each date are equal to the forward prices for crude oil that we observe in the financial markets. In addition, the discounted values of option payoffs calculated from the binomial lattice should equal the option prices that are observed in the financial markets. In the next step, one can calculate the value of the producing oil field at each node of the binomial tree by discounting the value of the cash flows that will be generated, given the oil prices that are realized in successive branches of the tree.

While this is somewhat more complicated than simply plugging in observed option prices from the financial derivatives markets for their real counterparts, it accomplishes the same thing. By calibrating the model, we are effectively using prices in the financial markets to determine real option values.

Real Option Valuation Formula

Because the Black-Scholes option pricing formula is so well known and can be programmed into handheld calculators, there is a temptation to apply the formula to real option problems. In most cases, the Black-Scholes formula is completely inappropriate for real option problems because it assumes that the options have a fixed maturity date, that they can be exercised only at the maturity date, and that the underlying investment has no cash payouts.

In this section, we provide an option pricing formula that we think is much more appropriate for valuing real options. This formula, which was developed by McDonald and Siegel (1986) based on earlier work by Samuelson and McKean (1965),[11] is based on similar assumptions as those of the Black-Scholes formula, except for the fact that it assumes that the underlying investment does have cash payouts and that the option can be exercised at any time and never matures. In other words, this is a formula for pricing an infinite-life American option.

The infinite-life option pricing model can be used to value real options in much the same way that the Black-Scholes formula is used to value European options. Although the formula looks complicated, it is simple to put into a spreadsheet and requires relatively few inputs. You can use it to calculate an initial value of a real option and perform sensitivity analysis to develop your intuition about what determines real option values.

Under these assumptions, the following equation describes the value of an American-style call option:

$$\text{Real Option Value} = (V^* - I)\left(\frac{V}{V^*}\right)^{\beta} \tag{3}$$

where each of the terms is defined as follows:

- V^* is the value of the underlying investment that "triggers" exercise of the real option. Note that this variable is an *output* of the model, rather than an input. The threshold value (V^*) is defined by

$$V^* = \frac{\beta}{(\beta - 1)} I \tag{4}$$

where

$$\beta = \frac{1}{2} - \frac{r_f - \delta}{\sigma^2} + \sqrt{\left(\frac{r_f - \delta}{\sigma^2} - \frac{1}{2}\right)^2 + \frac{2r_f}{\sigma^2}} \tag{5}$$

- I is the initial cost of making the investment. Hence, $V^* - I$ is the net present value that triggers the exercise of the real option. This difference is also sometimes referred to as the NPV hurdle, because it measures how high the NPV must be before the investment is initiated.
- V is the current value of the underlying investment.

[11] Robert L. McDonald and D. Siegel, "The Value of Waiting to Invest," *Quarterly Journal of Economics* 101, no. 4 (1986): 707–27, and Paul A. Samuelson and H. P. McKean Jr., "Rational Theory of Warrant Pricing," *Industrial Management Review* (1965).

Technical Insight
Comparing the Black-Scholes and Infinite-Life Real Option Pricing Models

Like the Black-Scholes formula, the infinite-life real option pricing model formula assumes that the value of the underlying asset changes randomly and that the volatility of the returns earned by the underlying investment and the interest rate are fixed. However, there are three important differences between the assumptions underlying this formula and the assumptions that are the basis of the Black-Scholes model:

- The first is that this model assumes that the option never expires, which should be contrasted to the Black-Scholes assumption that the option has a predetermined expiration date.
- The second difference is that the Black-Scholes formula assumes that the underlying investment pays no dividends. In contrast, the real option valuation formula assumes that the underlying investment generates cash flows that are proportional to the value of the investment and that this proportion does not change over time.
- The final difference is that the Black-Scholes formula assumes that the option is a European-style option that can be exercised only on the option's maturity date. In contrast, the real option valuation formula found in Equation 3 assumes that the option can be exercised at any time (is an American-style option) and that the option never expires.

- The risk-free rate of interest is r_f.
- The cash flow yield, or cash distribution as a fraction of the value of the investment, is represented by δ, which is assumed to remain constant over the life of the investment.
- σ is the standard deviation, or the *volatility* of the rate of return of the underlying investment.

We warned you that this formula looks complicated! However, as with the Black-Scholes model, we can learn from using this formula without understanding all of its mathematical intricacies. For example, from this formula, we can show that increased volatility increases real option values and, in addition, increases the NPV hurdle—that is, the minimum NPV required before the investment is triggered. Similarly, an increase in an investment's cash flow yield will decrease the option's value, because it implies that the investment's value appreciates at a slower rate.

Real Estate Development Option

To illustrate the use of this real option valuation model, we start with a simple example involving the purchase of 50,000 square feet of land that can be used to develop a 60,000-square-foot office building at a cost of \$10 million (I in Equation 3). It is currently not economical to initiate development, because the current value of such a building is only \$9 million (V in Equation 3). However, existing buildings in the area can be leased to yield (after taxes and all expenses) an 8% rate of return to the owner (δ in Equation 5). These buildings have volatilities (i.e., standard deviations—σ) in their annual rates of return of 10%. The risk-free rate, r_f, is assumed to equal 6%.

Note that if the property were developed today, it would have a value of only \$9 million but would cost \$10 million to develop, so the owner would lose money developing this property at the current time. However, the land has substantial value because it provides the owner with the *option to build* in the future. Substituting into Equation 3, we estimate that this property, which provides the owner with the option to build the above-described building, has a value of about \$287,667, i.e.,

$$\text{Real Option Value} = (V^* - I)\left(\frac{V}{V^*}\right)^{\beta}$$

$$= (\$11{,}732{,}501 - \$10{,}000{,}000)\left(\frac{\$9{,}000{,}000}{\$11{,}732{,}501}\right)^{6.772} = \$287{,}667$$

where β is calculated using Equation 5 as follows:

$$\beta = \frac{1}{2} - \frac{r_f - \delta}{\sigma^2} + \sqrt{\left(\frac{r_f - \delta}{\sigma^2} - \frac{1}{2}\right)^2 + \frac{2r_f}{\sigma^2}}$$

$$= \frac{1}{2} - \frac{.06 - .08}{.01} + \sqrt{\left(\frac{.06 - .08}{.01} - \frac{1}{2}\right)^2 + \frac{2 \times .06}{.01}} = 6.772$$

Consequently, as a speculator, you would be willing to pay up to \$287,667 for the right to develop this property.

The model also provides information about what we call the *trigger point*—in other words, how high the value must be before it makes sense to commence construction. Using Equation 4,

$$V^* = \left(\frac{\beta}{\beta - 1}\right)I = \left(\frac{6.772}{6.772 - 1}\right)\$10{,}000{,}000 = \$11{,}732{,}501$$

This calculation indicates that we should not begin construction until the value of the developed property exceeds \$11,732,501. Once the property value reaches this *trigger point,* the option to develop should be exercised.

How Do Changes in Model Parameters Affect Real Option Values?

It is useful to examine how the value of the option to develop the vacant land is affected by changes in the parameter estimates. For example, if the volatility parameter is increased to 15%, the option to develop the land, calculated using Equation 3, increases to \$669,762. If we reset the volatility to 10% but increase the dividend yield to 9%, the value of the option to develop the vacant land is reduced to \$191,225. Higher payouts make options less attractive relative to the underlying investments, because the owner of the building receives a payout on his or her investment, while the owner of the vacant land (i.e., the option holder) receives nothing.

Extensions of the Model

The real option valuation model in Equation 3 can be extended in a number of ways without too much difficulty. For example, suppose that the land is not completely vacant and is currently being used as a parking lot that generates income of \$100,000 per year. We can account for this complication by simply determining the value of the parking lot and adding this to the construction costs. For instance, in the earlier example, we assumed that the cost of constructing the building was \$10 million. To account for the loss of parking lot revenues valued at, say, \$1.8 million, we simply increase the development costs to \$11.8 million. In the same way, one can account for taxes on the land, which would generate a positive cost associated with holding the land.

Applying the Model to a Chemical Plant

We now apply the model to the valuation of the real options associated with building a manufacturing plant. Suppose that DuPont has an ethylene plant that currently produces 10,000 tons of ethylene per year. The plant is old and inefficient, but it still shows a fairly stable profit of \$1 million per year. DuPont is considering an opportunity to convert this facility to make diethelyne, which is a specialty chemical that will be substantially more profitable. DuPont's analysts estimate that the conversion will cost \$90 million and will generate an initial cash flow next year of \$10 million. The analysts expect these profits to increase 2% per year for the foreseeable future, and the standard deviation of the returns to the investment is estimated to be 15% per year. Under the assumption that these profit changes are random and that the discount rate is constant at 12% per year, we can estimate the value of the plant with a simple growth model, i.e.:

$$\begin{array}{c}\text{Value of the}\\ \text{Plant}\end{array} = \frac{\text{Cash Flow(Year 1)}}{(\text{Discount Rate} - \text{Growth Rate})} = \frac{\$10 \text{ million}}{.12 - .02} = \$100 \text{ million}$$

Under these assumptions, the investment has a value of \$100 million and a corresponding payout rate of 10%.

In order to evaluate this opportunity, one must also value the ethylene plant, because part of the cost of building the new plant is the opportunity cost associated with liquidating the old plant. We will assume that the old plant has a value of \$8.5 million, which is the capitalized value of the plant's cash flow at a 12% discount rate, i.e., $\$1 \text{ million} \div .12 = \$8.5 \text{ million}$. The total opportunity cost of building the new plant is equal to the \$90 million direct cost of converting the old plant to produce the new product plus the \$8.5 million opportunity cost associated with closing the old plant, for a total of \$98.5 million. Consequently, the new plant has a net present value of \$1.5 million, which equals the estimated value of the new plant (\$100 million) less the opportunity loss associated with closing the old plant and the cost of building the new plant (\$98.5 million), or NPV = \$100 million – \$98.5 million = \$1.5 million.

Although the investment has a positive NPV, the NPV is quite small compared to the value of the investment (i.e., 1.5%). Hence, it might be the case that the project would be worth more if DuPont delays the investment. To analyze this possibility, we value the project using the real option valuation formula found in Equation 3, with inputs as follows: The risk-free rate is 6%, the dividend yield is 10%, and the volatility of the underlying investment is 15%. Using Equation 3, we estimate that the chemical plant should not be converted until the new plant has a value of \$120,284,902. Under this conversion strategy, the option to convert has a value of \$7,857,267. That is,

$$\begin{aligned}\text{Real Option Value} &= (V^* - I)\left(\frac{V}{V^*}\right)^{\beta} \\ &= (\$120{,}284{,}902 - \$98{,}500{,}000)\left(\frac{\$100{,}000{,}000}{\$120{,}284{,}902}\right)^{5.5215} \\ &= \$7{,}857{,}267\end{aligned}$$

where β is calculated using Equation 5 as follows:

$$\beta = \frac{1}{2} - \frac{r_f - \delta}{\sigma^2} + \sqrt{\left(\frac{r_f - \delta}{\sigma^2} - \frac{1}{2}\right)^2 + \frac{2r_f}{\sigma^2}}$$

$$= \frac{1}{2} - \frac{.06 - .10}{.0225} + \sqrt{\left(\frac{.06 - .10}{.0225} - \frac{1}{2}\right)^2 + \frac{2 \times .10}{.0225}} = 5.5215$$

such that

$$V^* = \left(\frac{\beta}{\beta - 1}\right)I = \left(\frac{5.5215}{5.5215 - 1}\right)\$98{,}500{,}000 = \$120{,}284{,}902$$

Limitations of the Model

We need to emphasize that this model makes a number of strong assumptions that affect the accuracy of the value estimates that it provides. For example, the valuation model assumes that interest rates and volatilities are fixed for the life of the options. Clearly, that will not be the case in the real world. The model also requires payouts to be a fixed percentage of the investment's value and assumes that the investment opportunity lasts forever or never expires. However, as restrictive as this final assumption may sound, we can modify the model to accommodate deterioration in the value of the option due to the effects of competition by assuming that the value of the opportunity deteriorates at a fixed rate (say, 2% per year) over time. In essence, we assume that the growth rate in the value of the investment is negative.

It should also be noted that in the real estate and chemical plant examples considered in this section, no financial options correspond with the options that are being valued. Consequently, managers have much more leeway in how these investments can be valued than they would if their assumptions were constrained by the values that are observed in financial markets. Given this fact, it is imperative that when doing valuations like these, managers perform a sensitivity analysis that examines the extent to which the investment value is sensitive to the choice of the various parameters of the model.

6 Using Simulation to Value Switching Options

Three basic approaches are used to evaluate real options: the binomial lattice, the real option formula for options with an infinite maturity, and simulation. In this section, we illustrate the third approach.

Simulation is commonly used to solve very complex real option problems that involve multiple and interactive sources of uncertainty. To illustrate how simulation can be used, we will consider the valuation of an investment with a switching option, which provides managers with the flexibility to alter operations as economic conditions change. For example, in response to changes in demand for their products, managers generally have the flexibility to speed up or slow down production as well as to shift the mix of products they produce and sell. In addition, if the cost of inputs changes over time, managers can often switch to a lower-cost alternative input.

The type of *switching option* that we focus on in this section is the option to exchange one input for another—specifically, the option to switch between natural gas and fuel oil to fire an electric power plant. Another form of switching option is the option to exchange one output for another. For example, a toy-manufacturing plant that produces stuffed animal toys might switch between teddy bears and stuffed lions in response to perceived changes in demand. Designing a plant in a way that allows for such switching can be costly, so it is important to be able to value this option to switch.

Option to Switch Inputs

To illustrate the value derived from switching options, we consider the CalTex Power Company, which is considering the installation of a large electric power plant and has three alternative technologies under consideration: a natural gas–fired plant, a fuel oil–fired plant, and a co-fired or flexible plant that has the capability of switching between fuel oil and natural gas depending on which is the cheaper power source. The gas- and oil-fired plants each cost $50 million to build, while the flexible plant costs $55 million.

All three plants can produce the same amount of electricity and are expected to run at full capacity every year of their 10-year lives. Table 4 summarizes the expected revenues and costs of operating all three plant alternatives. To simplify the analysis, we assume the following:

- Plant revenues equal $32,550,000 in Year 1 and grow at a rate of 5% per year.
- All three plant technologies incur two costs of operations: fuel costs and fixed operating expenses.
- There are no costs associated with switching between the fuel alternatives with the flexible plant.[12] Moreover, the costs of the two fuels change only once per year, at the end of each year.
- The current price of natural gas is $7.75 per million Btu, and this cost is expected to rise at a rate of 2.5% per year. The price of fuel oil is currently $7.00 per million Btu, and it is expected to rise at a rate of 4.5% per year.[13]
- There are no income taxes.
- The plants are assumed to have no salvage values at the end of year 10.

The question that we will address is whether the added cost of the flexible power plant is worthwhile. Very simply, is the option to switch fuel sources worth the added $5 million it would cost?

Static NPV Analysis of the Plant Choices

Traditionally, DCF analysis calculates and discounts the cash flows that arise given the *expected* outcome scenario. We refer to this as **the static DCF analysis.** The results of the static analysis are presented in Table 4 for each of the three power plant alternatives. This analysis is static in that it assumes that the oil-versus-gas choice for the flexible plant is determined in advance based on the most likely scenario or, equivalently, the

[12] In practice, there are often setup or changeover costs that will deter a firm's willingness to switch from one mode of operation to another.

[13] Note that the prices of natural gas and fuel oil have been standardized in terms of the cost per million Btu of energy they are expected to produce. This allows us to compare the costs of the two fuel sources in terms of a common unit that can be related directly to the production of electric power.

Table 4 Static DCF Analysis of the Gas-Fired, Oil-Fired, and Flexible (Gas or Oil)-Fired Power Plants

	Expected Cost of Fuel/Btu[a]		Expected Fuel Costs[b]				Expected Free Cash Flow[d]		
Year	Gas	Fuel Oil	Gas	Fuel Oil	Other Expenses	Revenues[c]	Gas	Fuel Oil	Flexible[e]
0							$(50,000,000)	$(50,000,000)	$(55,000,000)
1	$7.94	$7.32	$(27,803,125)	$(25,602,500)	$(800,000)	$32,550,000	3,946,875	6,147,500	6,147,500
2	8.14	7.64	(28,498,203)	(26,754,613)	(800,000)	34,177,500	4,879,297	6,622,888	6,622,888
3	8.35	7.99	(29,210,658)	(27,958,570)	(800,000)	35,886,375	5,875,717	7,127,805	7,127,805
4	8.55	8.35	(29,940,925)	(29,216,706)	(800,000)	37,680,694	6,939,769	7,663,988	7,663,988
5	8.77	8.72	(30,689,448)	(30,531,457)	(800,000)	39,564,728	8,075,281	8,233,271	8,233,271
6	8.99	9.12	(31,456,684)	(31,905,373)	(800,000)	41,542,965	9,286,281	8,837,592	9,286,281
7	9.21	9.53	(32,243,101)	(33,341,115)	(800,000)	43,620,113	10,577,012	9,478,998	10,577,012
8	9.44	9.95	(33,049,179)	(34,841,465)	(800,000)	45,801,119	11,951,940	10,159,654	11,951,940
9	9.68	10.40	(33,875,408)	(36,409,331)	(800,000)	48,091,175	13,415,767	10,881,844	13,415,767
10	9.92	10.87	(34,722,293)	(38,047,751)	(800,000)	50,495,733	14,973,440	11,647,983	14,973,440

	Gas	Oil	Flexible
NPV[f]	$ (503,157)	$462,277	$(528,142)
IRR	9.81%	10.19%	9.81%

[a]The expected cost of fuel per Btu for gas and fuel oil is estimated from the current costs and the expected rate of inflation. For example, the current cost of natural gas is $7.75 per Btu, and the anticipated rate of inflation is 2.5%. Thus, the expected price of gas in one year is equal to $7.94/Btu = $7.75/Btu × (1 + .025).

[b]The expected cost of fuel equals the product of the total fuel (measured in Btus) consumed by the plant to produce 500,000 kWh of electricity times the fuel cost per Btu from either gas or fuel oil. We calculate the total fuel required to produce electricity using the heat rate for the plant, which can differ if the plant is being run with natural gas. However, we assume a heat rate of 7.0 for gas and oil here. The heat rate equals the ratio of total fuel consumed in Btus divided by the total kWh of electricity produced.

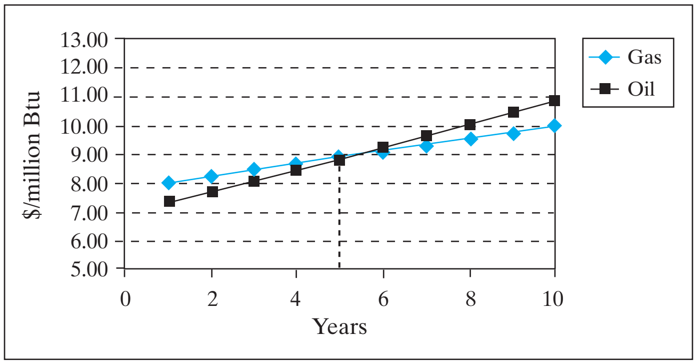
[c]Plant revenues are the same for each of the three plant alternatives and are expected to grow at a rate of 5% per year.

[d]Free Cash Flow = Revenues – Expected Fuel Cost – Other Expenses. We assume that there are no taxes or salvage value for any of the plant alternatives.

[e]The fuel cost incorporated into the calculation of the expected free cash flow for the flexible plant is equal to the lower of the expected fuel cost using natural gas or fuel oil for each year.

[f]Net present value is calculated using a 10% cost of capital for each plant alternative.

Figure 7 **Expected Costs of Natural Gas and Fuel Oil**



expected cost of the fuel at each future date. As we illustrate in Figure 7, the expected cost of oil is less than that of gas for Years 1 through 5, but then rises above it, triggering a switch from fuel oil to natural gas in Year 6.

Although the cash flows from these plants are not necessarily equally risky and the risk of the switching plant may change over time as the preferred fuel changes, we will initially value these plants assuming a constant 10% discount rate. As we show in Table 4 with this discount and the assumed cash flows, the fuel oil plant is the preferred alternative and the flexible plant has a negative NPV. Based on this static analysis, it appears that the flexible plant does not offer sufficient expected fuel cost savings to warrant spending an additional $5 million.

Our static NPV analysis, however, ignores an important benefit associated with the flexible plant: It assumes that the fuel input is predetermined based on the expected cost of the alternative fuels and thus ignores the possibility that management can alter its choice of fuel throughout the life of the flexible plant, using gas in those years in which gas is less expensive and oil in those years in which oil offers the better alternative. Because the costs of natural gas and fuel oil are uncertain, there is a benefit to having the flexibility to respond to future market conditions that cannot be known at the time the analysis is being done. As we demonstrate below, the value of this flexibility can be estimated with a simulation model that incorporates the dynamic effects of having the option to switch fuel sources so as to minimize the cost of fuel.

Dynamic Analysis—Valuing the Option to Switch Fuel Sources

In our static analysis, we established that the fuel oil–fired plant is preferred to the natural gas–fired plant. Now we must determine whether adding the option to switch between fuel oil and natural gas using the flexible plant is worthwhile. Consequently, we want to value the fuel cost savings that would accrue to the owner of the flexible plant when compared to those of the fuel oil–fired plant.

Modeling future oil and gas prices Although this problem is somewhat more difficult to solve than the problems that we have described up to now, the overall process is very similar. The basic idea is to specify a price process for the commodities that are the value drivers for the power plant projects. In this case, we have two value drivers—the prices of natural gas and fuel oil per Btu—so we must specify a price process for both commodities. By this we mean the process that describes their volatilities and how these volatilities change over time. Up to now, we have assumed that commodities are like stocks and follow a random walk, which means that price changes in any given time period are independent of the previous price changes. This assumption simplifies option pricing problems, but it is not realistic for most commodities. Commodity prices are generally assumed to be mean-reverting, which means that if prices increase a lot in one year, they are likely to fall in the following year. In the Technical Insight box titled "Modeling Natural Gas Prices Using Geometric Brownian Motion and Mean Reversion," we will provide more detail about the price processes that analysts use to describe oil, gas, and other commodity prices.

Did you know?

Origins of Brownian Motion

Studying the random behavior of pollen particles suspended in water, Scottish scientist Robert Brown is credited with identifying the phenomenon we now refer to as Brownian motion. Many years later, Albert Einstein developed the mathematical properties of Brownian motion that are commonly used today.

However, Louis Bachelier (1870–1946), in his 1900 dissertation *Theorie de la Spéculation* (and in his subsequent work, especially in 1906, 1913), described the ideas behind the random walk of financial market prices, Brownian motion, and martingales (before Einstein!) Sadly, his innovativeness was not appreciated until it was rediscovered in the 1960s.

(*http://cepa.newschool.edu/het/profiles/bachelier.htm*)

We will also use observed prices from the financial markets to value the option to switch between natural gas and fuel oil, but in this case, we cannot simply plug in the prices of traded derivatives. Instead, we must first specify a price process for natural gas and fuel oil and then calibrate the process so that it generates values for observed forward and option prices that are consistent with the prices observed in the financial markets. We will not provide the technical details of such a calibration process, but will assume that from such a process we arrive at the following set of parameter estimates:

	Rate of Reversion in Prices*	**Mean-Reversion Price = Forward Price****	**Standard Deviation of Price Changes**
Natural gas	1.0	$7.75/Btu	30%
Fuel oil	0.6	$7.00/Btu	20%

*Historical estimates of the rate of reversion to the mean for natural gas have been faster than for oil prices.

**This is also the initial price of each commodity in the current period (Year 0).

We have one final attribute that we need to estimate. Note that because we have two commodity prices (natural gas and fuel oil) that must be considered when valuing the switching option, it is also necessary that we estimate the correlation between the prices of natural gas and fuel oil per Btu. If the prices of natural gas and fuel oil are very highly correlated, the option to switch will be less valuable. Intuitively, the gain from switching is much less if fuel oil prices are likely to be high whenever natural gas prices are high. Because fuel oil and natural gas are frequent substitutes, high demand for one is likely to put pressures on the price of the other, which does make the two commodities positively correlated. However, the correlation is not particularly high, and for our purposes, we assume a positive correlation of 0.25.

Did you know?

What's a Wiener Process?

The increment to Brownian motion defined by Equation 7 is often referred to as a *Wiener process,* named after the American mathematician Norbert Wiener (1894–1964).

We must again note that because the simulated price process has been calibrated using derivative prices, the resulting prices that we estimate using Equation 6 are not the actual expected prices, but instead are *risk-adjusted* expected prices or certainty-equivalent prices. Consequently, the expected cash flows that we calculate from this *calibrated price process* are certainty-equivalent cash flows that can be valued by discounting them at the risk-free rate.

Using simulation to value the flexibility option At this point, we have made all the assumptions and forecasts we need to characterize the future prices of natural gas and fuel oil and are ready to value the option to switch between the two sources of fuel. To solve for the value of this flexibility option, we use the simulation process. This process requires the following three steps.

Step 1: Identify the sources of uncertainty, characterize the uncertainty using an appropriate probability distribution, and estimate the parameters of each distribution. In this instance, the sources of uncertainty associated with the value of the option to switch fuel sources are the costs of natural gas and fuel oil for each of the next 10 years. We have modeled the prices of natural gas and fuel oil per Btu using Equation 6.

Step 2: Define the summary measure that we are attempting to estimate. In this instance, our objective is to evaluate the present value of the expected fuel cost savings for Years 1 through 10 if the flexible plant is constructed.

Step 3: Run the simulation. We use 10,000 iterations, or trials, in the simulation of fuel prices. The iterations begin with the cost of fuel per Btu observed in Year 0. Then the change in price is simulated using Equation 7 (one for natural gas and one for fuel oil) and added to the cost/Btu for Year 0 to get the simulated price for Year 1. The process is repeated to obtain fuel prices for Years 2 through 10. The simulated prices for gas and oil form a single price path for each of the fuels. We repeat this process a total of 10,000 times to form 10,000 price paths for oil and, likewise, 10,000 price paths for natural gas. These simulated costs of gas and oil per Btu are then used to calculate the annual fuel cost savings that accrue to the flexible plant alternative. Because we are analyzing the value of the flexible plant compared to that of the fuel oil plant alternative, the flexibility option creates savings only when the cost of natural gas falls below the price of fuel oil.

Once we have identified the distributions of fuel cost savings from the simulation, we calculate the value of these expected annual savings by discounting them using the risk-free rate of 6%.

The column titled "Average Annual Fuel Savings" in Panel a of Table 5 contains the average annual cash flow savings that are expected to accrue to the flexible plant when it always selects the lower-cost fuel source. Discounting these savings back to the present using the risk-free rate of interest produces a value of $22,252,213, which, when compared to the $5 million added cost of building the flexible plant, provides an NPV of $17,252,213. Based on this analysis, CalTex should select the flexible plant technology.

Panel b of Table 5 contains a histogram of the values generated from each of the simulations for the flexibility option. We will not engage in a full sensitivity analysis using the simulated distribution here but will point out that the discounted value of the simulated cash flows associated with the switching option is at least equal to the $5 million additional investment in 94.88% of the simulation trials.

Technical Insight

Modeling Natural Gas Prices Using Geometric Brownian Motion and Mean Reversion

Brownian motion without mean reversion has been compared to the path that a drunk might take home after a night out on the town. We can think of the path marked by the staggering steps of the drunken man as he wanders toward home as a representation of Brownian motion. Based on this analogy, Brownian motion is sometimes referred to as a random walk.

Now consider the possibility that the drunken man takes his faithful dog, Sparky, to the bar. Because Sparky doesn't drink, he can lead his master home. With Sparky along to guide him, the man staggers along aimlessly until the dog's leash tugs him back toward the direct path taken by Sparky. The path of the drunken man as he staggers along behind Sparky's lead provides a rough approximation of Brownian motion with mean reversion. The leash, like mean reversion, keeps the drunken man from wandering too far from the path home.

Stock prices are generally modeled as Brownian motion without mean reversion—that is, as a random walk. However, the prices of a commodity, such as natural gas, are generally modeled as Brownian motion with mean reversion, which means that if prices get too far away from a long-term equilibrium price, they tend to gravitate back towards that price, which is usually governed by the cost of production and the level of demand.

We can mathematically express this price process in Equation 6, which defines the price of natural gas at the end of Year 1, $P_{Gas}(1)$, as the sum of the observed price of natural gas today, $P_{Gas}(0)$, plus the change in the price of gas over the coming year, $\Delta P_{Gas}(1)$:

$$P_{Gas}(1) = P_{Gas}(0) + \Delta P_{Gas}(1) \tag{6}$$

Next, for a mean-reverting geometric Brownian motion process, $\Delta P_{Gas}(1)$ can be defined by the following equation:

$$\Delta P_{Gas}(1) = P_{Gas}(0)[\text{Predictable Component } (\mu) + \text{Unpredictable Component } (\sigma_{Gas}\,\varepsilon)]$$

$$\Delta P_{Gas}(1) = P_{Gas}(0)[\mu + \sigma_{Gas}\,\varepsilon] \tag{7}$$

where the predictable component $\mu = \alpha_{Gas}[\ln(L) - \ln(P_{Gas}(0))]$ and α_{Gas} is the rate at which gas prices revert back to the mean price, L_{Gas}. For example, a reversion rate of two would suggest that prices would revert to the mean price in half a period (six months in our example). The unpredictable component is described by ε, which is a normal random variable with mean zero and unit standard deviation, and σ_{Gas}, which represents the volatility of the logarithm of natural gas price changes. The unpredictable component is the source of randomness in the price path.

Substituting Equation 7 into Equation 6, we see that the price of gas next year is

$$P_{Gas}(1) = P_{Gas}(0)[1 + \alpha_{Gas}[\ln(L) - \ln(P_{Gas}(0))] + \sigma_{Gas}\varepsilon] \tag{8}$$

COMMON VALUATION MISTAKES

Valuing Real Options

Because of the difficulties associated with valuing real options, it is no surprise that it is very easy to make mistakes in the valuation process. Some of the more common mistakes are discussed below

Trying to Fit the Problem into the Black-Scholes Model

There is a natural tendency to use financial formulas even when the underlying assumptions upon which the models were developed are not consistent with the realities of the situation. This tendency has led many analysts to "force-fit" the Black-Scholes option pricing formula to real option problems in cases where the model is not even remotely appropriate.

The Black-Scholes model is derived using a set of assumptions, but a particularly onerous assumption is that the option can be exercised only at maturity (i.e., it is a European option). In fact, real options should generally be viewed as American options that can be exercised at any time. Still another issue with the Black-Scholes model is that it assumes that there are no intermediate cash flows from the underlying asset. This is so even though, in most cases, the underlying investment generates cash flows, and these cash flows, like dividends on a share of stock, affect the value of the investment and consequently the value of any options based on the investment's value. Finally, the exercise price of real options, unlike the fixed exercise price of the Black-Scholes model, is typically unknown.

Using the Wrong Volatility

Analysts often use the wrong volatility when valuing natural resource investments with embedded real options. For example, when valuing options on natural resource investments, the volatility of the price of the underlying commodity is erroneously used rather than the volatility in the value of the appropriate underlying asset. For example, as we described earlier, the volatility of the value of a producing oil field is the volatility that should be used to value a lease that gives the holder the option to extract oil. However, in practice, it is difficult to determine this volatility, so analysts sometimes take the shortcut of using the volatility of oil prices. This shortcut would produce the correct answer if oil prices were like stock prices and followed a random walk (that is, price changes are serially uncorrelated). However, oil prices, like most commodity prices, tend to be mean-reverting. This means that a very large positive price change is more likely to be followed by a drop in price than by another increase. When this is the case, the volatility of commodity prices will be greater than

7 SUMMARY

The growth in the use of real option analysis over the last decade can largely be attributed to its broad-based appeal as a tool to communicate and estimate the value that flexibility can add to real investments. The examples in this chapter illustrate that there can be a substantial increment to project value whenever managers have the flexibility to act opportunistically when managing real investments in an uncertain environment.

Surveys of business practice indicate that MBA programs have been very successful in ingraining the use of static discounted cash flow analysis into hoards of newly minted MBAs. However, this success has had an unfortunate side effect: Static DCF analysis has supplanted some of the commonsense business analyses used in years past that recognized the importance of operational flexibility. This has meant that the early adherents of a more quantitative approach to project analysis based on static DCF analysis may have underestimated the extent to which flexibility is a source of value. The computational power of computer spreadsheets, simulation software, and specialized option the volatility of investments that generate cash flows that depend on the commodity prices. This will, in turn, lead to an overvaluation of real options.

Assuming That the Exercise Price of the Real Option is Fixed

Unlike financial or contractual options, the exercise price for a real option is generally not fixed. For example, your firm may have the option to undertake a project today at a cost of $8 million or to delay the investment for another year. However, if over the year the price of the project's output increases significantly, it is likely that the cost of initiating the project will also have risen. In fact, as a general rule, the costs of initiating a project are higher when the project is more valuable. For example, when housing prices increase, we often see a building boom, which drives up the costs of lumber and other building materials. In general, a high correlation between an investment's value and the cost of implementing the investment has the effect of lowering the value of the option to invest. So ignoring this factor leads to overvaluation of the real option.

Overestimating the Value of Flexibility

The fundamental source of the value of real options comes from optimally taking advantage of flexibility. However, to capture this value, the firm's management must be both willing and able to exercise the options when conditions warrant doing so. In reality, managers tend not to ruthlessly exercise real options, contrary to what the theory suggests. For example, management may fail to act to shut down a facility when it is optimal to do so because of loyalties to employees, suppliers, or customers. Furthermore, even absent conflicting motives, management may not have the incentives or the ability to exercise real options in an optimal fashion. The message here is that one should not place much value on options that can be exercised *in theory* but are unlikely to be exercised *in reality*.

Double-Counting Risk

In our analysis of the value of embedded options in oil fields, we used forward prices to calculate certainty-equivalent cash flows and discounted these cash flows at the risk-free rate of interest. This analysis *correctly* associates a risk-free discount rate to certainty-equivalent cash flows. However, industry applications sometimes mistakenly use forward prices to represent expected prices, which are then used to calculate what are erroneously *assumed* to be expected cash flows, which are then discounted at a risk-adjusted rate. Hence, the analyst ends up adjusting both the cash flows and the discount rate for risk, which results in a conservative estimate of value.

valuation software on the desks of analysts today makes real option analysis a practical reality. However, as we learned with the adoption of quantitative techniques such as static DCF, change takes time, so it may be years before the real option approach to project valuation is as prevalent as static DCF. But we believe that this is the direction of the future.

In this chapter, we presented three approaches that can be used to value investments with real options:

- The first approach we discussed was the binomial lattice approach. This and related approaches are used extensively on Wall Street to value financial options, and because of its success there, it has been adopted in some settings to value real investments.
- The second approach was the application of a real option formula, which to our knowledge is not really used in industry. We presented this formula because, like the Black-Scholes formula, it is simple to program and provides a rough estimate of real option values. Like the Black-Scholes formula, this formula provides estimates that are a bit too simple and too rough to use for a final valuation of a major investment.

Table 5 **Dynamic Analysis of the Option to Switch Fuels—Comparing the Fuel Oil–Fired Plant and the Flexible Plant**

Panel a. Certain Equivalent Expected Annual Fuel Cost Savings

Year		Average Annual Fuel Savings
1		2,518,650
2		3,215,166
3		3,180,391
4		3,094,741
5		3,138,945
6		3,121,089
7		3,038,148
8		3,012,766
9		3,001,547
10		2,972,366
Risk-free rate	6.0%	
Value switching option (fuel savings)	$ 22,252,213	
Less: Cost of option	(5,000,000)	
NPV (switching option)	$ 17,252,213	

These savings correspond to the benefits of having the option to switch between fuel sources compared to the fuel oil–fired plant. The latter was determined to be the better alternative when the natural gas and fuel oil–fired plants were compared.

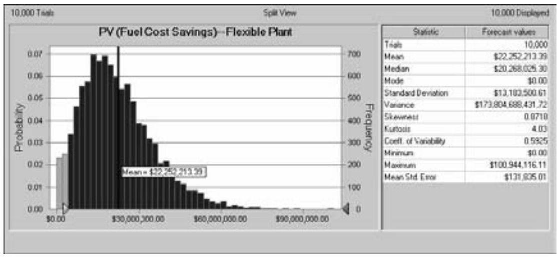
However, it may provide a useful starting point to determine whether a more refined analysis is warranted.

- The final tool that we discussed was simulation analysis. This is becoming increasingly more important for valuing investments, and we think that this is the tool of the future. Simulation is capable of handling the most complicated investments in a way that does not require a great deal of technical sophistication from the analyst. In reality, major investments often contain many sources of managerial flexibility as well as a number of sources of uncertainty, and simulation is the only available tool that is capable of dealing with this level of complexity.

The application of real option analysis varies across industries and even firms within industries. For example, real option analysis is used extensively in the energy and natural resource industries, where commodity markets are active and derivative prices are readily observable. We hasten to point out, however, that although the complexity of the analysis presented in this chapter is greater than that in prior chapters, the real option valuation approaches that are actually used in practice can be even more complicated. However, because these industry approaches are generally based on either the simulation approach or the binomial approach described in this chapter, an understanding of

Table 5 ***continued***

Panel b. Distribution of Flexible Option Values (Crystal Ball Output)

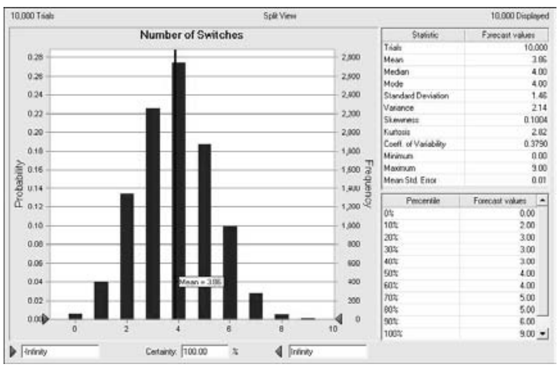


PV(Fuel Cost Savings) = present value of the cost of reduced fuel consumption due to switching fuel to a low-cost source.

Fuel cost savings = the difference in the total annual fuel cost of fuel oil and natural gas when natural gas is cheaper than fuel oil.

Certainty = Percent of the 10,000 iterations when the present value of the fuel cost savings exceeds the $5 million added cost of the flexible plant.

Panel c. Distribution of the Number of Switches over 10 Years (Crystal Ball Output)



Definitions:

- A fuel switch occurs whenever the cost of fuel can be lowered by changing from one fuel to another.
- The number of switches equals the number of times within the 10-year plant life.

this material provides the foundation for understanding the most complicated valuation approaches used in industry.

In other industries, real options are evaluated with approaches that are less sophisticated than those described in this chapter. For example, although flexibility plays a very important role in real estate development, real estate investors typically do not use explicit valuation models to value real estate development options. In addition, while managers do understand the value of timing options (which exist for almost any major investment project), they tend to use relatively simple and intuitive approaches for valuing the option to wait as well as the option to abandon.

Why are real option valuation approaches more popular in energy and natural resource industries than in other industries? In these industries, the payoffs between the real options and the options that are traded in the financial markets are quite close, allowing managers to value the investment without having to use a lot of judgment. In other cases, such as the valuation of land with development opportunities, there are no financial option prices that are directly analogous to the development option being valued. In these cases, real option analysis is viewed less favorably, because the manager must substitute his or her judgment about the future distribution of the relevant values for the financial option prices that do not exist. While the need to exercise judgment in a valuation problem is not unique to the application of real option models, it is natural for managers to be more reluctant to exercise judgment in the application of tools they are less familiar with, and thus to be less confident about their judgment. As we have emphasized earlier, in those cases where more managerial judgment is required, it is important that the valuation analysis also include a detailed sensitivity analysis.

In the next chapter, we evaluate investment strategies, which we will define as the potential for a *series* of investment opportunities. For example, when Toyota considers an assembly plant to build pickup trucks in Texas, it is considering a single investment. However, when the company considers whether it makes sense to assemble cars in Eastern Europe, where the company does not currently build automobiles, it is evaluating an investment strategy that can eventually lead to assembly plants in multiple locations for a variety of cars and trucks. The fact that both flexibility and uncertainty are inherent in all investment strategies implies that some sort of option analysis is required for a disciplined approach to evaluating the strategy.

PROBLEMS

1 ANALYZING AN OIL LEASE AS AN OPTION TO DRILL FOR OIL Suppose you own the option to extract 1,000 barrels of oil from public land over the next two years. You are deciding whether to extract the oil immediately, allowing you to sell the oil for $20 per barrel, or to wait until next year to extract the oil and sell it then for an uncertain price. The extraction costs are $17 per barrel. The forward price is $20, and you know that oil prices next year will be either $15 per barrel or $25 per barrel, depending on demand conditions. Are you better off extracting the oil today or waiting one year? Explain how your answer might be different if prices next year are either more or less certain but have the same mean.

2 CONCEPTUAL ANALYSIS OF REAL OPTIONS Huntsman Chemical is a relatively small chemical company located in Port Arthur, Texas. The firm's management is contemplating its first international investment, which involves the construction of a petrochemical plant in São Paulo, Brazil. The proposed plant will have the capacity to produce 100,000 tons of the plastic pellets that are used to manufacture soft drink bottles. In addition, the plant can be converted over to produce the pellets used in the manufacture of opaque plastic containers such as milk containers.

The initial plant will cost $50 million to build, but its capacity can later be doubled at a cost of $30 million, should the economics warrant it. The plant can be financed with a $40 million nonrecourse loan provided by a consortium of banks and guaranteed by the Export Import Bank. Huntsman's management is enthusiastic about the project, as its analysts think that the Brazilian economy is likely to grow into the foreseeable future. This growth, in turn, may offer Huntsman Chemical many additional opportunities in the future as the company becomes better known in the region.

Based on a traditional discounted cash flow analysis, Huntsman's analysts estimate that the project has a modest NPV of about $5 million. However, when Huntsman's executive committee members review the proposal, they express concern about the risk of the venture, based primarily on their view that the Brazilian economy is very uncertain. Toward the close of their deliberations, the company CEO turns to the senior financial analyst and asks him whether he has considered something the CEO has recently read about called "real options" in performing his discounted cash flow estimate of the project's NPV.

Assume the role of the senior analyst and provide your boss with a brief discussion of the various options that may be embedded in this project, and very roughly sketch out how these options can add to the value of the project. (*Hint:* No computations are required.)

3 OPTION TO ABANDON Newport Mining has a lease with two years remaining, in which it can extract copper ore on a remote island in Indonesia. The company has completed the exploration phase and estimates that the mine contains 5 million pounds of ore that can be extracted. The ore deposit is particularly rich and contains 37.5% pure copper.

Newport can contract with a local mining company to develop the property in the coming year at a cost of $1.2 million. Three-fourths of the cost of development must be paid immediately and the remainder at the end of one year. Once the site is developed, Newport can contract with a mine operator to extract the ore for a cash payment equal to $0.60 per pound of ore processed or $1.60 per pound of copper produced.[14] The total cost must be paid in advance at the beginning of the second year of operations. This amounts to a cash payment in one year of $3 million.

At the end of one year, Newport can contract to sell the copper ore for the prevailing spot price at that time. However, because the spot price at the end of the year is unknown today, the proceeds from the sale of the refined copper are uncertain.

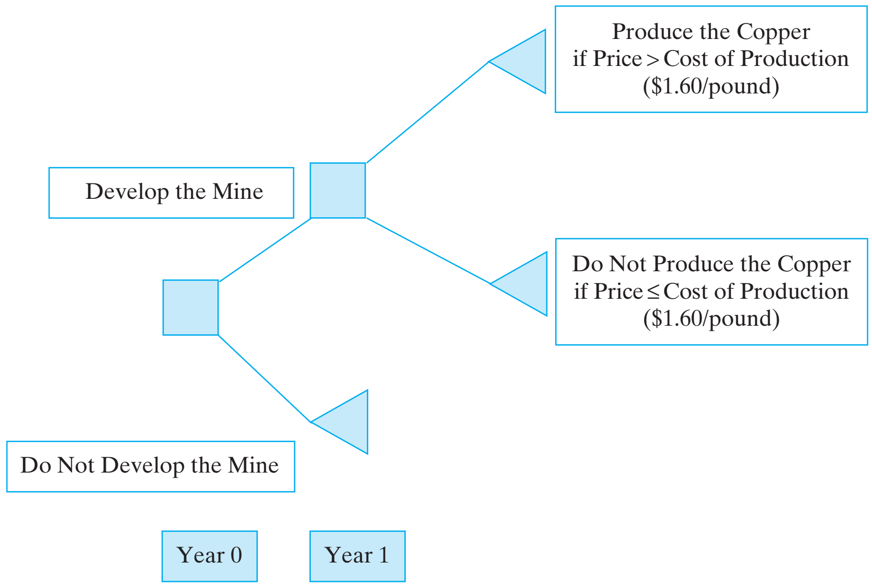
The current price is $2.20 per pound, and commodity analysts estimate that it will be $2.50 a pound at year-end. However, because the price of copper is highly volatile, industry analysts have estimated that it might be as high as $2.80 or as low as $1.20 per pound by the end of the year. The price of copper is expected to stay at $2.80 or as low as $1.20 throughout the second year. As an alternative to selling the copper at the end-of-year spot price, Newport could sell the production today for the forward price of $2.31 and eliminate completely the uncertainty surrounding the future price of copper. However, this strategy would require that the firm commit today to producing the copper. This, in turn, means that Newport's management would forfeit the option to shut down the plant should the price be less than the cost of producing the copper.

Given the risk inherent in exploration, Newport requires a rate of return of 25% for investments at the exploration stage but requires only 15% for investments at the development stage. The risk-free rate of interest is currently only 5%.

- **a.** What is the expected NPV for the project if Newport commits itself to the development, extraction, and sale of the copper today and sells the copper in the forward market?
- **b.** What is the NPV of the project if the production is *not* sold forward and Newport subjects itself to the uncertainties of the copper market?
- **c.** Using the decision tree, construct a diagram that describes Newport's payoff from the investment that includes the option to extract the ore at the end of one year.
- **d.** What is the lease worth to Newport if it exercises its option to abandon the project at the end of Year 1? Should the firm proceed with the development today?
- **e.** If Newport decided to extract the ore itself, how could it use the copper call options to hedge the risk of mining for the copper? The price of a European call option on one pound of copper with an exercise price of $1.68 and maturity of two years is $0.70.

[14] Because the ore contains 37.5% copper and there are 5 million pounds of ore in the mine, the total is 1.875 million pounds of copper to be produced at a cost equal to $0.60 × 5 million pounds of ore, or $3.0 million. Thus, the cost per pound to produce the ore is $3.0 million/1.875 million pounds, or $1.60 per pound.

Newport's Options to Develop and Produce a Copper Mine



4 VALUING AN AMERICAN OPTION J&B Drilling Company has recently acquired a lease to drill for natural gas in a remote region of southwest Louisiana and southeast Texas. The area has long been known for oil and gas production, and the company is optimistic about the prospects of the lease. The lease contract has a three-year life and allows J&B to begin exploration at any time up until the end of the three-year term.

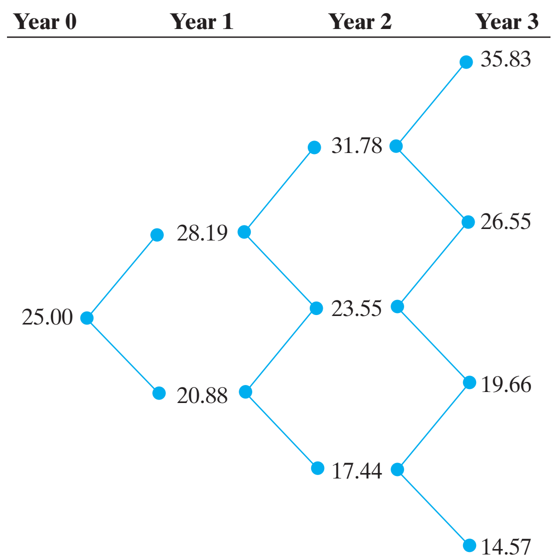
J&B's engineers have estimated the volume of natural gas they hope to extract from the leasehold and have placed a value of $25 million on it, on the condition that explorations begin immediately. The cost of developing the property is estimated to be $23 million (regardless of when the property is developed over the next three years). Based on historical volatilities in the returns of similar investments and other relevant information, J&B's analysts have estimated that the value of the investment opportunity will evolve over the next three years as shown in the figure on next page.

The risk-free rate of interest is currently 5%, and the risk-neutral probability of an uptick in the value of the investment is estimated to be 46.26%.

a. Evaluate the value of the leasehold as an American call option. What is the lease worth today?

b. As one of J&B's analysts, what is your recommendation as to when the company should begin drilling?

5 OPTION TO SWITCH INPUTS—CRYSTAL BALL EXERCISE The Central and Southeast Power Company of Mobile, Alabama, is considering a new power plant that will allow it to switch between gas and oil. The company has a contract to provide it with gas for $8 that is sufficient to produce one unit of electric power. (The numbers are standardized to one unit for ease of computation.) However, the plant can also be run using fuel oil. The price of fuel oil is uncertain, and the firm's analysts believe that the uncertain future price can be characterized as a triangular distribution with a minimum value of $2, a



most likely value of $7, and a maximum value of $12. Next year, the plant is expected to produce one standard unit of electrical power that can be sold for $10.

a. What is the expected cash flow from the power plant for next year, if the cost of fuel is set equal to the minimum of the expected costs of gas and oil? (*Hint:* There are no taxes, and the only expenses that the plant incurs are for fuel.)
b. Construct a simple Crystal Ball simulation model for the plant's cash flow using a triangular distribution for the cost of fuel oil and selecting the minimum-cost source of fuel to run the plant. What is the expected cash flow from the power plant based on your simulation?
c. If the cost of capital for the plant is 10% and the cash flows for the plant are expected to be constant forever, what is the value of the plant?

6 REFINERY WITH THE OPTION TO SWITCH OUTPUTS—CRYSTAL BALL EXERCISE The Windsor Oil Company is considering the construction of a new refinery that can process 12 million barrels of oil per year for a period of five years. The cost of constructing the refinery is $2 billion, and it will be depreciated over five years toward a zero salvage value. The plant can produce either gasoline or jet fuel, and the product can be changed once each year based on the prices of the two products. The refinery can convert 42 gallons (per barrel) of crude into 90% of this quantity of gasoline, or 70% if jet fuel is produced. The residual can be sold, but for no net profit.

Windsor's analysts characterize the distribution of gasoline prices for next year and each of the next five years using a triangular distribution that has a minimum value of $1.75 per gallon, a most likely value of $2.50, and a maximum of $4.00. They characterize the price per gallon of jet fuel using a triangular distribution with a minimum value of $2.50 per gallon, a most likely value of $3.25, and a maximum value of $5.00 a gallon. The price of crude is fixed via forward contracts over the next five years at $25 per barrel. Windsor also estimates that the cost of refining is equal to 35% of the selling price of the particular product being produced.

Windsor faces a 30% tax rate on its income and uses a cost of capital of 10% to analyze refinery investments. The risk-free rate is 5.5%.

a. What is the NPV of the refinery if it produces only jet fuel (because the revenues under this alternative are higher, based on the most likely price)?
b. Construct a simulation model for the refinery, with the triangular distribution described above, that makes the price of gasoline and jet fuel random variables. What is the NPV of the refinery investment if the firm selects the higher-valued product to produce, based on the realized prices of gasoline and jet fuel?

7 CONCEPTUAL ANALYSIS OF REAL OPTIONS Highland Properties owns two adjacent four-unit apartment buildings that are both on 20,000 square feet of land near downtown Portland, Oregon. One of the properties is in very good condition, and the apartments can be rented for $2,000 per month. The units in the other property require some refurbishing and in their current condition can be rented for only about $1,500 per month.

Recent zoning changes, combined with changes in market demand, suggest that both lots can be redeveloped. If they are redeveloped, the existing units would be torn down and new luxury apartment buildings would be built on the site, each with 10 apartment units. The cost of the 10-unit buildings is estimated to be about $1.5 million, and each of the 10 apartment units can be rented for $2,500 per month under current market conditions. Similar properties that have been refurbished are selling for 10 times their annual rentals.

a. Identify the real option(s) in this example.
b. What are the basic elements of the option(s) (i.e., the underlying asset on which the option is based, the expiration date, and the exercise price)?
c. Estimate the value of the option to develop the property. (*Hint:* Make any assumptions you must to arrive at an estimate.)

8 CONCEPTUAL ANALYSIS OF REAL OPTIONS The destruction that Hurricane Katrina brought to the Gulf Coast in 2005 devastated the city of New Orleans as well as the Mississippi Gulf Coast. Notably, the burgeoning casino gambling industry along the Mississippi coast was nearly destroyed. CGC Corporation owns one of the oldest casinos in the Biloxi area, and it was not destroyed by Katrina's tidal surge because it is located several blocks off the beach. Because of the near-total destruction of many of the gambling properties located along the beach, CGC is considering the opportunity to make a major renovation in its casino. The renovation would transform the casino from a second-tier operation into one of the top attractions along the Mississippi Gulf Coast. The question that the firm faces involves placing a value on the opportunity to renovate the property.

CGC's analysts estimate that it would cost $50 million to do the renovation. However, based on the uncertainties associated with the redevelopment of the region, the firm's financial analyst estimated that the casino, under the current conditions, would be valued at only $45 million. Alternatively, CGC could continue to operate the casino, in which case it expects to realize an annual rate of return of 10% on the value of the investment. Moreover, the estimated return of 10% is highly uncertain. In fact, the volatility (standard deviation) in this rate of return is probably on the order of 20%, while the risk-free rate of interest is only 5%.

a. What is the NPV of renovation of the property if undertaken immediately?
b. What is the value of having the option to renovate in the future? (*Hint:* You can assume that the option never expires.)

APPENDIX

Constructing Binomial Lattices

The binomial option pricing model utilizes a binomial tree to specify the evolution of the value of the underlying asset through time. The tree begins with the current value of the underlying asset equal to S_0. In one period, the value is equal to either uS_0 or dS_0. Moreover, to ease the computational problem involved in evaluating the tree when there are many periods, it is now standard practice to restrict the price changes in the tree to be symmetrical, such that u equals $1/d$. This means that in the next period, the possible values are reduced to three: u^2S_0, udS_0, and d^2S_0. This version of the binomial tree is called a *recombining binomial tree* or a *binomial lattice.*

When the risk-neutral valuation approach is used to value options, we construct what is referred to as a risk-neutral binomial lattice, in which the expected return to the underlying asset is the risk-free rate, r, but the asset's volatility, σ, is the same as that of the risky asset.

To illustrate how u and d are determined, let's first assume that the asset is risk free. In this setting, the value of the asset appreciates at the risk-free rate less any payout over the period. For example, the value of a share of stock will appreciate at the risk-free rate less the dividend yield, δ. So, if the stock were worth \$10.00 at the beginning of the period, the risk-free rate is 6%, and the dividend yield is 4%, then the value of the stock at the end of one period will be $\$10e^{.06-.04} = \$10.20$.[15] Thus, in the absence of uncertainty, we can now define u as follows:

$$S_1 = S_0 e^{r-\delta}$$

Now we introduce uncertainty into S_1 using the annualized volatility of the returns to the asset, σ:

$$uS_0 = S_0 e^{r-\delta} e^{\sigma} = S_0 e^{r-\delta+\sigma} \quad \text{and} \quad dS_0 = S_0 e^{r-\delta-\sigma}$$

Multiple Time Steps per Year

To this point, we have assumed that one period equals one year; however, this need not be the case. If, for example, one period equals one month, then the risk-free rate, payout, and annual volatility have to be adjusted to reflect one-twelfth of a year; i.e., the monthly volatility would be s 2 $\frac{1}{12}$, and S_1 would become

$$uS_0 = S_0 e^{(r-\delta)\left(\frac{1}{12}\right)+\sigma\sqrt{\frac{1}{12}}} \quad \text{and} \quad dS_0 = S_0 e^{(r-\delta)\left(\frac{1}{12}\right)-\sigma\sqrt{\frac{1}{12}}}$$

[15] For a commodity such as oil held in the tanks of a refinery or cattle in the feed lots of a meatpacker, the payout is referred to as a *convenience yield,* because owning the physical commodity has value to the holder (e.g., it meets an inventory need) that is lost with the passage of time.

More generally, if there are n time steps per year such that each *time step* is of length $1/n$th of a year, we define the value of the underlying asset at the end of one time step as follows:

$$uS_0 = S_0 e^{(r-\delta)\left(\frac{1}{n}\right)+\sigma\sqrt{\frac{1}{n}}} \quad \text{and} \quad dS_0 = S_0 e^{(r-\delta)\left(\frac{1}{n}\right)-\sigma\sqrt{\frac{1}{n}}}$$

Building the Binomial Lattice for the National Petroleum Oil Field Investment

To implement the binomial option pricing model to value the National Petroleum Company's investment opportunity, we must first determine the future values of the drilling opportunity specified in the binomial lattice. We will specify these values using the procedure just discussed. To do so, we need the following information:

a. The starting point for the value of the oil field investment opportunity today is $42.034 million.
b. The risk-free rate of interest (r) is 6%.
c. The convenience yield (δ) is 7%.
d. The standard deviation (volatility) of the annual returns to the drilling investment (σ) is assumed to be .15.

Given this information, we can calculate the binomial distribution of values for the oil development project at the end of the first year as follows:

$$\text{High Value} = S_0 u \quad \text{where} \quad u = e^{r-\delta+\sigma}$$

or

$$\text{Low Value} = S_0 d \quad \text{where} \quad d = e^{r-\delta-\sigma}$$

Substituting for the determinants of u and d, we calculate

$$u = e^{r-\delta+\sigma} = e^{.06-.07+.15} = 1.1503 \text{ and } d = .852$$

such that the two possible values for National's drilling opportunity at the end of Year 1 are

$$S_0 u = \$42.03 \text{ million} \times 1.1503 = \$48.35 \text{ million}$$

and

$$S_0 d = \$42.03 \text{ million} \times .852 = \$35.82 \text{ million}$$

At the end of Year 2, there are three possible valuations for the drilling opportunity, corresponding to the following sequence of events:

$$S_0 uu = \$42.03 \text{ million} \times 1.1503 \times 1.1503 = \$55.62 \text{ million}$$

$$S_0 ud = S_0 du = \$42.03 \text{ million} \times 1.1503 \times .852 = \$41.20 \text{ million}$$

$$S_0 dd = \$42.03 \text{ million} \times .852 \times .852 = \$30.51 \text{ million}$$

In Year 3, the number of nodes expands to four, corresponding to $S_0 uuu$, $S_0 uud$, $S_0 udd$, and $S_0 ddd$. Figure 6 contains the completed binomial lattice for the National Petroleum drilling opportunity.